Corrigendum to Eduardo Montero and Dean Yang, "Religious Festivals and Economic Development: Evidence from the Timing of Mexican Saint Day Festivals," *American Economic Review* 2022, 112(10): 3176-3214.

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Abstract

Montero and Yang (2022) study the long-run economic impact of the timing of Catholic saint day festivals in Mexico. We show that "agriculturally coinciding" festivals (those coinciding with peak planting or harvest months) have negative impacts on the long-run economic development of localities. This paper provides revised tables, figures, and empirical analyses of Montero and Yang (2022) to rectify data coding errors discovered by Campbell et al. (2024b). Key coefficient estimates in the revised analyses are consistent in sign, remain statistically significant at conventional levels, and yield the same overall conclusions as the original paper. We also address other comments raised by Campbell et al. (2024b). Errors in the identification of municipalities' patron saints should simply give rise to classical measurement error, attenuating our coefficient estimates. Many of the analyses conducted by Campbell et al. (2024b) are not properly described as "robustness tests". Some are versions of analyses we included in the original paper to examine mechanisms underlying the effects, while others show interesting evidence of heterogeneity in the effects of agriculturally-coinciding festivals. Analysis of robustness to excluding high-leverage observations is redundant as we provide already provide such analysis in our original paper.

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1. Introduction

Societies worldwide spend substantial resources celebrating religious festivals. How do festivals influence economic and social outcomes? In Montero and Yang (2022), we study the impact of Catholic patron saint day festivals in Mexico, exploiting two features of the setting: (i) municipal festival dates vary across the calendar and were determined in the early history of towns after Spanish conquest, and (ii) there is considerable variation in the intra-annual timing of agricultural seasons. We compare municipalities with "agriculturally-coinciding" festivals (those that coincide with peak planting or harvest months) to other municipalities, examining differences in long-run economic development and social outcomes. We find that agriculturally-coinciding festivals have negative effects on household income and other development outcomes. They also lead to lower agricultural productivity and higher share of the labor force in agriculture, consistent with agriculturally-coinciding festivals inhibiting the structural transformation of the economy. Agriculturally-coinciding festivals also lead to higher religiosity and social capital, potentially explaining why such festivals persist in spite of their negative growth consequences.

A project to examine the robustness and reproducibility of results of 17 non-experimental papers in the *American Economic Review* has been conducted by Campbell et al. (2024a). Our paper, Montero and Yang (2022), is one of the 17 examined papers, and is also the subject of a separate "robustness report", Campbell et al. (2024b).

We prepared this corrigendum in response to Campbell et al. (2024b). The bulk of the corrigendum provides revised tables, figures, and empirical analyses of Montero and Yang (2022) to rectify data coding errors discovered by Campbell et al. (2024b). We thank the authors of Campbell et al. (2024b) for pointing out these coding errors. Importantly, we provide corrections for *all* tables, figures, and empirical analyses of Montero and Yang (2022). By contrast, Campbell et al. (2024b) show only a subset of corrected analyses – for just one of two primary outcome variables, and for a selected subset of potential empirical specifications.

In our full set of revised analyses, key coefficient estimates are consistent in sign with the original Montero and Yang (2022) estimates, remain statistically significant at conventional levels, and yield the same overall conclusions as the original paper. In some select cases the coefficients show more differences with respect to the original paper (while remaining of the same sign and statistical significance), and these are the ones on which Campbell et al. (2024b) focus. We would argue that it is important to assess whether the original conclusions of the paper hold up when viewing the full set of analyses. By focusing on corrections to just a subset of selected analyses, Campbell et al. (2024b) paint an incomplete picture.

We also address other comments raised by Campbell et al. (2024b). We find that many of their key critiques are misplaced. Any errors in identification of a municipality's patron saint are likely to simply represent classical measurement error and result in attenuation bias; our results should thus be seen as lower bounds of the true effects. The use of official Vatican-prescribed festival dates (rather than municipalities' actual celebrated dates) is not an error, but is intentional and key to causal identification. Their analyses of changes in coefficient estimates when perturbing planting- and harvest-festival date windows are versions of our original analyses; these help reveal underlying mechanisms, and should not be considered "robustness tests". Evidence on how treatment effects differ nonlinearly across municipalities with different levels of maize suitability are also not "robustness tests", but rather analysis of treatment effect heterogeneity.

Campbell et al. (2024b) state that coefficient estimates are "robust at a 5% confidence level in 24% of the robustness checks we ran." This 24% figure is not a valid statistic, since – as we have discussed – many of the analyses conducted by Campbell et al. (2024b) are not actually "robustness tests". Many are actually analyses of treatment effect heterogeneity, or are versions of analyses we already conducted in the original paper aimed at revealing underlying mechanisms behind effects. Other analyses they include in their set of regressions – such as those including local saints in the sample – are conceptually suspect (as we discuss below) and also are versions of analyses we already conducted in the original paper. The analysis of high-leverage observations is redundant as we already provide such analysis, using the approach of Young (2019).

For easy reference, our section numbers below map to the sections of the Campbell et al. (2024b) robustness report.

2. Data and Coding

2.1. *Code Review*

Campbell et al. (2024b) find two types of errors in our data processing. First, codes for missing household income data (coded as 99999998 or 99999999 in the Census microdata) were averaged with non-missing household income data to calculate mean household income at the municipality level. These missing codes should have been set to missing, rather than taken as income values, before the municipality-level mean was calculated.

Second, there were errors in the calculation of a municipality's festival overlap with its optimal harvest dates. The error occurred when a festival and the optimal agricultural harvest dates occur in adjacent months on opposite sides of the change in calendar year (one occurring in December and the other in January). In such cases, our calculation incorrectly calculated the date distance in the same calendar year (yielding numbers greater than 300), when we should have calculated date distance between December and January dates of successive calendar years (yielding numbers between 0-60, a range within which "agriculturally coinciding" festivals may occur). This error occurred for eight municipalities.

These are genuine errors, for which we apologize. We thank Campbell et al. (2024b) for their careful review of our code that surfaced these errors.

After correcting these errors, the revised share of municipalities with agriculturally-coinciding festivals is as follows. In the former New Spain region of Mexico, 4.49 percent of municipalities have planting-coinciding festivals, and 7.38 percent have harvest-coinciding festivals, for a total of 11.88 percent having agriculturally-coinciding festivals. These figures compare to (respectively) 4.49, 7.20, and 11.69 in the original paper.[1]

In this corrigendum, we use the corrected data and revise all tables, figures, and empirical analyses, including those in the Online Appendix. For ease of comparison, we have numbered all these items as they are originally numbered in Montero and Yang (2022).

2.1.1. Results with Corrected Data: Analyses in Main Text

In our full set of revised analyses, all coefficient estimates are consistent in sign with the original Montero and Yang (2022) estimates, remain statistically significant at conventional levels, and yield the same overall conclusions as the original paper.

We now review the main analyses of the paper in detail, discussing how estimates change from the original to the corrected analyses.

Table 1 provides a test for manipulation of festival dates to avoid planting or harvest overlap. Estimates are nearly unchanged compared to the original estimates. Our original conclusion remains: there is no evidence that localities choose patron saints to avoid overlap of festival dates with harvest or planting months.

Figure 4 shows results of statistical tests of differences in fixed or historical characteristics of municipalities, comparing those with and without agriculturally-coinciding festivals. The general pattern from the original paper remains: there are no statistically significant differences in municipality characteristics that raise omitted variable concerns or provide alternate explanations for our findings. Echoing findings from the original paper, there are two characteristics for which the tests shows imbalance at the 10% level of significance. Municipalities with coinciding festivals are more likely to be suitable for maize and less likely to have experienced drought in 1545. The conclusion from our original discussion in Montero and Yang (2022) (p. 3196) remains valid: "These

[1]In Mexico overall, the corresponding corrected percentages are 6.31 percent, 8.28 percent, and 14.59 percent. In the original paper, the respective percentages were 6.31, 7.94, and 14.25 percent.

preexisting differences should *improve* the long-run economic development prospects of places with agriculturally coinciding festivals, suggesting that our estimates may be lower bounds (in absolute value) of the true (negative) effects of coinciding festivals on economic development. In subsequent regression tables we show estimates without and with these and other controls. Results are always robust to their inclusion or exclusion."

Table 2 is our primary results table, presenting coefficient estimates of the effect of agriculturally-coinciding festivals on our two primary economic development outcomes: log household income and the index of economic development. For completeness, it is important to examine both primary outcome variables, rather than just log household income as Campbell et al. (2024b) do.

The conclusions from our original paper remain the same with the revised Table 2 analyses. Agriculturally-coinciding festivals lead to significantly lower levels of economic development, whether measured by log mean household income or by the index of economic development. Point estimates are robust to the set of controls and fixed effects included on the right hand side of regressions. Compared to the original estimates, point estimates in the revised Table 2 regressions with corrected data are slightly smaller in magnitude, but remain statistically significant at conventional levels. In the most inclusive specification, column 5, coinciding festivals lead to 13.2 log points lower household income, compared to 20.6 log points in the original paper (p-values are 0.064 with robust standard errors, and 0.024 with Conley (1999) standard errors). The revised analysis finds that coinciding festivals lower the index of economic development by 0.11 standard deviations, compared to the original estimate of 0.13; p-values are 0.059 with robust standard errors, and 0.009 with Conley (1999) standard errors.

Figure 5 examines the components of the index of economic development, to explore what is driving effects on the index. The corrected figure is very similar to the figure in the original paper. For nearly all component variables, the coefficient is negative and statistically significantly different from zero. Coinciding festivals lead to lower household income, literacy, employment, and education. The negative effects of coinciding festivals reveal themselves in a variety of measures of economic development.

Figure 6 tests whether the negative effects of coinciding festivals are indeed concentrated in the months closest to harvest and planting. The figure tests this prediction by presenting coefficient estimates of impacts of festivals on log household income in other months relative to planting and harvest months. The pattern of results in the revised Figure 6 is very similar to the original figure. We quote the original text from Montero and Yang (2022) (p. 3201-3202), as it remains fully valid: "The largest effects occur when festivals coincide exactly with the period prior to the planting month or following the harvest month. The planting-month and harvest-month effects are both statistically significantly different from zero. Effects of festivals in other months adjacent to planting or harvest are much closer to zero, and none are statistically significant. These results are consistent with the hypothesis that festivals tighten liquidity constraints and crowd out key investments in the planting and harvest periods."

Table 3 examines impacts of coinciding festivals on agricultural productivity and structural transformation of the economy. Coefficient estimates are very similar to the original estimates in terms of magnitudes and statistical significance levels. The corrected estimates of Table 3 yield the same conclusion as the original paper: coinciding festivals lead to lower agricultural productivity and less transition of employment from agriculture to the modern sectors of the economy.

Figure 7 displays regression estimates examining when in history the effect on economic development emerged. The dependent variable is population density, a proxy for economic development commonly used in historical studies. The pattern of results in the corrected figure is very similar to the original paper. The conclusions remain the same: there are no effects of coinciding festivals on population density in 1570 or 1650, and negative effects emerge for 1900 and persist in 2010. Negative effects of coinciding festivals emerged historically some time between 1650 and 1900.

The last set of results in the main body of the paper is **Table 4**, which examines effects of coinciding festivals on religiosity, social capital, and income inequality. Columns 1 and 2 of the table (regressions for religiosity and social capital) are unaffected by the data correction, because the regressions are run on a subset of 131 municipalities in the LAPOP data, none of which were affected by the coding error determining agriculturally-coinciding festivals. Column 3 of the table, for which inequality (IQR of earned incomes) is the dependent variable, is affected by the data coding errors. The correction yields very little change in the column 3's coefficient estimate. The corrected estimate reveals that coinciding festivals lead to lower income inequality, as in the original paper.

2.1.2. Results with Corrected Data: Analyses in Online Appendix

In this corrigendum, we also provide corrected figures, tables, and regression results for all items in Online Appendix of Montero and Yang (2022). We provide a higher-level overview here.

Appendix B provides corrected maps of festival months, maize planting months, maize harvest months, days between festivals and optimal planting dates, days between festivals and optimal harvest dates, coincidence of festivals with optimal planting months, and coincidence of festivals with optimal harvest months, for New Spain as well as all of

Mexico.[2] The maps are very similar to those in the original paper.

Figure C1 validates the FAO GAEZ data on harvest month with actual production data. Corrected analyses provide very similar validation of the data, as in the original paper.

Figure C2 conducts a randomization inference exercise, simulating the data to examine the distribution of placebo treatment effects that would emerge with randomly assigned placebo coinciding festivals (Young, 2019). The conclusions remain the same as in the original paper: when using corrected data, the estimated t-statistic of our coinciding-festival coefficient is much more negative than the majority of placebo festival assignments. The randomization inference p-value is 0.0065 (compared to 0.0004 in the original paper). The conclusion remains the same as in the original paper: our results are unlikely to be driven by outliers or high-leverage observations.

Figure C3 displays coefficient estimates on coinciding festivals in regressions for each individual component of the religiosity and social capital indexes. Corrected results are very similar to the original estimates. Coinciding festivals have positive effects on each of the religiosity index and social capital index components, with the exception of political group participation.

Figure C4 displays coefficient estimates when varying the 30-day windows defining the planting and harvest periods, in 15-day increments up to 60 days before and after the optimal planting and harvest dates. The pattern in the corrected figure is very similar to the pattern in the original one. We quote here the original discussion in the Online Appendix of Montero and Yang (2022), as it remains valid: "The estimates suggest an interesting time dimension to the impacts of festivals on development. First, the negative estimated effects of festivals coinciding with planting appear for various rolling windows prior to planting but converge toward zero following planting. Second, we observe the opposite timing for harvest festivals: the negative estimated effects of festivals coinciding with harvest only begin to appear following harvest (and are statistically insignificant and close to zero prior to harvest). Additionally, the estimates show that the main results are not particularly sensitive to the specific 30-day window we consider. These results are consistent with the hypotheses that the timing of festivals is important for understanding their development consequences, and that festivals can crowd out investments and decrease development when they occur in times when time-sensitive economic opportunities exist."

The remainder of the Online Appendix contains an assortment of additional supplementary tables, which we discuss briefly here.

[2]Note for all of Mexico maps: the maps in this corrigendum are now regenerated using SF-based workflow (no FORTIFY/SP::GSIMPLIFY). This resolves minor geometry warping in the all-Mexico figures (especially in the north) caused by the deprecated SP::GSIMPLIFY + FORTIFY join code.

- **Table D1** tests for manipulation of festival dates to avoid planting or harvest overlap in the sample of all Mexican municipalities (not just for New Spain). Estimates are nearly unchanged, and the conclusion is the same: there is no evidence that festival dates intentionally avoid harvest or planting months.

- **Table D2** shows the actual regression coefficients underlying **Figure 4** in the main text, which we discussed above. **Table D3** does the same for the sample of all Mexican municipalities; balance in municipality characteristics is maintained in the corrected analyses.

- **Table D4** shows estimates of effects of harvest-coinciding and planting-coinciding festivals separately, on the two primary outcome variables. As in the original results, all coefficients are negative for both harvest- and planting-coinciding festivals, in all regressions where the dependent variable is either log household income or the index of economic development. As in Table 2, coefficients are slightly lower than the original estimates. Even in the original estimates, not all coefficients in this table were statistically significantly different from zero at conventional levels. Among those that were statistically significantly different from zero in the original table, the coefficients on harvest-coinciding festivals in column (5) of Panels A and B have become statistically insignificant after correction.

- **Table D5** presents coefficient estimates when estimating regressions for log household income and for the index of economic development, but for the sample of all Mexican municipalities (not just for New Spain). Corrected estimates reveal a similar pattern as in the original paper. Coefficients are negative in all regressions for both outcomes, but consistently smaller in magnitude compared to Table 2's results for New Spain municipalities. In column 5 (with all controls), the coefficient on coinciding festivals is no longer statistically significant at conventional levels in either panel. As we say in the original paper (p. 3202), "This reduction in coefficient magnitudes is consistent with festivals having less impact outside of New Spain for the reasons discussed in Section IIIB."

- **Table D6** tests whether municipalities with missing dates have similar geographic characteristics to other municipalities. As in the original paper, this appears to be true.

- **Table D7** shows that municipalities with local saints have similar characteristics to other municipalities. This remains true in the corrected analyses.

- In **Table D8** and **Table D9**, we show the main results are very similar when we classify municipalities with missing festival dates either as having coinciding or noncoinciding festivals.
- **Table D10** presents the main results when including municipalities with local saints in the sample. In the corrected analyses, as in the original analyses, inclusion of municipalities with local saints does not have a major effect on the estimates.
- **Table D11** presents regression estimates of the effect of coinciding festivals on migration outcomes. The corrected coefficient estimates are identical to those in the original paper (to the third decimal place as presented in the table): coinciding festivals lead to higher shares of population reporting having been in a different country five years before.

2.2. Review of Original Sources

A key data contribution of Montero and Yang 2022 is the dataset we created of saint day festival dates for Mexican municipalities (nearly 1,600 in New Spain, and about 2,300 in all of Mexico). The first step involved identifying the patron saint celebrated by each municipality. The second step was determining the official celebration date of the patron saint.

In Montero and Yang (2022) (p. 3187-3188), we describe the process we followed in taking the first step: "First, we determined the patron saint for each municipality. An online data source, the *Encyclopedia of Municipalities in Mexico* INAFED (1988), provided information on the patron saint for 77.85 percent of municipalities. For the remaining municipalities, we (i) directly contacted municipalities by phone (13.97 percent of municipalities) and (ii) conducted online searches, requiring at least two sources for each municipality (7.49 percent). In the phone calls, we focused on contacting municipal government officials, local churches, or schools. Overall, we were able to determine a patron saint for 99.3 percent of municipalities in Mexico." Our Online Appendix A.1 provides extensive further detail, including source information for municipalities not covered in INAFED (1988).

For a select subset of 40 municipalities, Campbell et al. (2024b) review our stated sources and critique decisions on patron saints and celebration dates. They conduct this review for 20 municipalities in the state of Guanajuato and 20 in the state of Oaxaca. In each of the two states, they selected the first 20 in alphabetical order for their review.

We thank Campbell et al. (2024b) for their data-checking exercise. Below we respond to key elements of their critique.

2.2.1. *Non-Random Selection of Sample for Data-Checking Exercise*

A basic first point is that a data-checking exercise of this sort – while very valuable – should have been done on a randomly-selected sample of municipalities. An assessment based on a sample from just two states, that was not randomly selected even within those states, cannot be claimed to be representative of the full sample of municipalities.

2.2.2. *Disagreement with Identity of Patron Saint*

In some of the cases they reviewed, they identify a different patron saint for a municipality than we identified. It is certainly true that in some cases the sources are ambiguous. Any large data collection exercise of this sort that requires review of a wide variety of sources (including phone calls to municipality institutions in a minority of cases) will yield differing decisions in some cases.

To the extent that there are errors in the identification of patron saints, this should likely simply be considered a source of classical measurement error. It is well known that classical measurement error in a right-hand-side variable of a regression leads to attenuation bias (Griliches and Hausman, 1986); our estimates of the effect of coinciding festivals are therefore likely attenuated (biased towards zero). We should thus take the estimates in our paper as lower bounds of the true effects.[3]

2.2.3. *Official Vatican-Prescribed Dates*

Campbell et al. (2024b) make much of the fact that in some cases the date celebrated by a municipality departs from the official Vatican-prescribed festival date for the patron saint. Given that actual celebrated dates depart sometimes from the actual date, they criticize our choice to focus on the official than the actual celebration dates in our analyses.

But our choice to focus on the official Vatican-prescribed date is intentional, and is a key element of our identification strategy. In other words, it is a feature, not a bug. In our original paper, p. 3178, we give an overview of our approach to causal identification: "The main identification assumption of our cross-sectional analysis is that the coincidence of a locality's festival date and the timing of its agricultural seasons is uncorrelated with omitted variables that may also affect long-run economic development. We take three approaches to bolstering the credibility of the identifying assumption. First, we review historical sources on the determination of saints by Mexican municipalities, and find no evidence contradicting our identification assumption. Second, in determining whether

[3] Campbell et al. (2024b) do acknowledge that the main consequence of errors in the identification of patron saints would be attenuation of coefficient estimates (p. 7). They then go on to speculate that the errors "may not be random", but provide no further discussion or evidence that the errors could be other than random, classical measurement error.

a municipality has a coinciding festival, we make two key analytical choices: (i) we use only "official" saint celebration dates set by the Catholic Church in the Vatican, rather than potentially endogenous actual celebration dates chosen by municipalities, and (ii) we determine optimal planting and harvest periods using a global database of agricultural conditions, rather than a municipality's actual planting and harvest dates (which may also be endogenous). Third, we present a number of empirical tests to confirm the plausibility of the identification assumption, particularly after controlling for key fixed effects and exploiting only cross-town variation within Mexican states."

Focusing on the official Vatican-prescribed date allows us to avoid bias due to localities endogenously choosing to celebrate festivals away from the Vatican-prescribed dates.[4] For example, a worry would be that localities with better local institutions, higher education levels, or other features promoting development might choose to celebrate their festivals on dates away from the official Vatican-prescribed dates, so as to not have agriculturally-coinciding festivals. Municipalities that remain with agriculturally-coinciding festivals would be negatively selected, imparting a negative bias to our estimated coefficient on agriculturally-coinciding festivals.

It is important to avoid such selection bias. We therefore focus only on official Vatican-prescribed dates in defining whether a municipality has an agriculturally-coinciding festival. This approach is conceptually akin to an instrumental variables context where the official Vatican-prescribed date serves an instrument for the actual celebrated date. We are unable to take an IV approach because the data on actual celebrated dates is incomplete, so our approach is like a "reduced form" regression in an IV context in which we regress the second-stage outcome variable on the instrument itself.

In practice, when the data on the actual celebration date is available, we find that there is typically not very much difference between the actual celebration date and the official date. From footnote 16 of the original paper: "Most municipalities in our sample (84 percent) celebrate their saint day festival on exactly the date officially prescribed by the Vatican or other Roman Catholic authority outside of Mexico. Among those that do deviate from the official date, the median number of days of divergence from the official date is seven."

In sum, the point of Campbell et al. (2024b) that festivals are often celebrated on dates that depart from their official dates is a misplaced critique. Using the actual celebrated dates to determine whether a festival is "agriculturally coinciding" would contaminate the key right-hand-side variable in our analysis, as it would include "bad" (endogenous)

[4]From our original paper, p. 3188: "Once we had determined the patron saint for (nearly all) municipalities, we then sought to ascertain the official (Vatican-prescribed) festival celebration date for that saint. We focus on the official timing rather than the date each municipality actually celebrates their festival, as the latter might be endogenous to local conditions and outcomes."

variation in festival celebration dates, raising questions about whether the coefficient on the indicator for agriculturally-coinciding festivals can be interpreted as a causal effect. We avoid this "bad" variation in festival dates by focusing on official dates prescribed by the Vatican, as these official dates cannot be endogenous to decisions made on the ground in Mexican municipalities.

2.2.4. Other Festivals

Another critique by Campbell et al. (2024b) is that they identified other festivals that municipalities claim are important for them, other than their patron saint-day festival. Key examples are national celebrations (celebrated by all of Mexico) such as Christmas, *Semana Santa* (Holy Week, which includes Easter), and the festival of the Virgin of Guadalupe.

This critique is also misplaced. Our focus is on estimating the effect of patron saint-day festivals. The fact that other festivals or religious celebrations exist is not an issue, as long as they are not omitted variables of concern (i.e., as long as they are conditionally uncorrelated with the indicator for "agriculturally-coinciding festivals"). Our inclusion of a rich set of fixed effects – in particular fixed effects for the calendar week of the festival – and the robustness of our estimates to their inclusion indicates that national-level festivals such as Christmas, Easter, and the like are not likely sources of omitted variable bias.

3. Replication

3.1. Regression Model

In all their analyses of robustness, Campbell et al. (2024b) focus on the regression specification of Table 2, Panel A, column (5). This is the regression where the dependent variable is log household income, and the full set of controls and fixed effects are included on the right-hand-side.

This approach is incomplete. First of all, we have two primary outcome variables that we examined side by side in Montero and Yang (2022): log household income and the index of economic development. In addition, it is important to examine robustness of results to varying the set of controls in the regression, to gauge whether there is any evidence of omitted variable bias.

In this corrigendum, we provide revised analyses using corrected data for all regressions in the original Montero and Yang (2022) paper. As in the original paper, we always examine regressions for both primary outcome variables (log household income and the index of economic development) and for regression specifications with an increasingly inclusive set of controls and fixed effects on the right-hand-side.

3.2. Computational Reproduction

Campbell et al. (2024b) ran a "computational reproduction" of our analyses, testing whether they were able to replicate our results using the data and code we provided in our replication package to the *American Economic Review*.[5] They state that "for all the regression tables in this paper, and, using the author-provided data and code, were able to reproduce the original results exactly."

We thank the authors of Campbell et al. (2024b) for conducting this computational replication and for confirming they were able to reproduce our empirical estimates.

3.3. Robustness Results

In their Table 1, Campbell et al. (2024b) estimate a regression corresponding to Table 2, Panel A, column 5 of Montero and Yang (2022), but with corrected data. They estimate a coefficient on the indicator for agriculturally-coinciding festivals of -0.134.

We are unable to replicate this coefficient exactly. Instead, we find a coefficient of -0.132. While we cannot replicate the corrected coefficient exactly, we confirm their finding that the coefficient in that regression falls by about a third compared to the original regression, while remaining statistically significantly different from zero at conventional levels (whether using robust standard errors or Conley (1999) standard errors accounting for spatial correlation).

We present the full set of corrected coefficient estimates in this corrigendum. Regressions in Table 2, Panel B present results for the second of our two primary outcome variables, the index of economic development. In column 5 (the regression with the full set of controls), the corrected coefficient is -0.438, compared to the original coefficient of -0.537. The corrected coefficient in this regression also remains statistically significantly different from zero at conventional levels with either type of standard error.

In the final column of their Table 1, Campbell et al. (2024b) estimate an additional regression that adds municipalities with "local saints" to the sample. They provide no rationale for adding these municipalities to the sample. They simply add this regression to the table and report its coefficient (-0.123, not substantially different from the estimate without adding the local saint municipalities).

Adding municipalities with local saints to the sample requires some discussion. In Montero and Yang (2022) (p. 3188), we note that some saints are "local" in that they are not recognized by Vatican calendars or celebrated outside of that municipality in Mexico. Less than 2% of municipalities have local saints. Our main analyses exclude such local-saint municipalities due to concerns about endogeneity. The concern is similar to the

[5]The replication package can be found here: <https://www.openicpsr.org/openicpsr/project/161801/version/V2/view>.

concern (discussed above in Section 2.2.3 of this corrigendum) about using municipalities' endogenously-selected actual celebration dates rather than official Vatican-prescribed dates: inclusion of local saints in the sample may lead to bias if municipalities with distinctive characteristics are more likely to have local saints and choose their local-saint celebration dates with an eye towards whether they overlap with the harvest or planting seasons.

In our original paper, as well as the corrected analyses of this corrigendum, we exclude municipalities with local saints from the primary sample as the timing of their festivals is potentially endogenous. In Appendix Table D7, we test whether local-saint municipalities have different characteristics than other municipalities, finding no statistically significant differences. Additionally, we present regression results showing that results are robust to the inclusion of local-saint municipalities in the sample (see our Appendix Table D10).

3.3.1. Harvest vs. Planting and Rolling the Festival Date Window

Campbell et al. (2024b) present analyses in which they perturb the 30-day harvest and planting festival windows. They perturb ("roll") the 30-day windows by moving them 3, 5, 7, and 10 days *later* for planting festivals and 3, 5, 7, and 10 days *earlier* for harvest festivals. In these analyses, they focus on regression specifications that separately estimate effects of two indicator variables, for festivals overlapping with planting months and festivals overlapping with harvest months. Regressions with these four different ways of perturbing festival month windows have coefficient estimates that are smaller in magnitude and not statistically significantly different from zero. Campbell et al. (2024b) conclude that our results are not robust: "results are sensitive to the exact cutoff date chosen."

This criticism is frankly somewhat strange, since these analyses of Campbell et al. (2024b) are versions of analyses that we already conducted in our original paper. In fact, we made the same point in the original paper: coefficient estimates indeed become smaller in magnitude when making exactly these kinds of perturbations of the 30-day windows defining planting- and harvest-coinciding festivals.

We conducted these analyses in the original paper in a systematic investigation to understand more precisely the specific windows in which festivals occurring near planting and harvest periods have the largest negative effects. We investigated how estimates of the effect of planting- and harvest-coinciding festivals vary when perturbing the windows in 30-day shifts (Figure 6 of the main text) and in 5-day shifts (Appendix Figure C4). In Sections 2.1.1 and 2.1.2 above, we discuss that these analyses with corrected data yield the same conclusions as in the original paper.

We quote here the original discussion in Montero and Yang (2022) (Online Appendix p. 23-24), as it remains valid: "The estimates suggest an interesting time dimension to the impacts of festivals on development. First, the negative estimated effects of festivals coinciding with planting appear for various rolling windows prior to planting but converge toward zero following planting. Second, we observe the opposite timing for harvest festivals: the negative estimated effects of festivals coinciding with harvest only begin to appear following harvest (and are statistically insignificant and close to zero prior to harvest)."

We point out these patterns because they help reveal potential underlying mechanisms behind the effects of agriculturally-coinciding festivals. The fact that effects of planting-coinciding festivals are largest for those occurring *prior to* planting suggests that key mechanisms behind the negative effects are interference with investments and labor time for activities that occur prior to planting, such as land clearing and preparation, and purchases of fertilizer and other inputs.

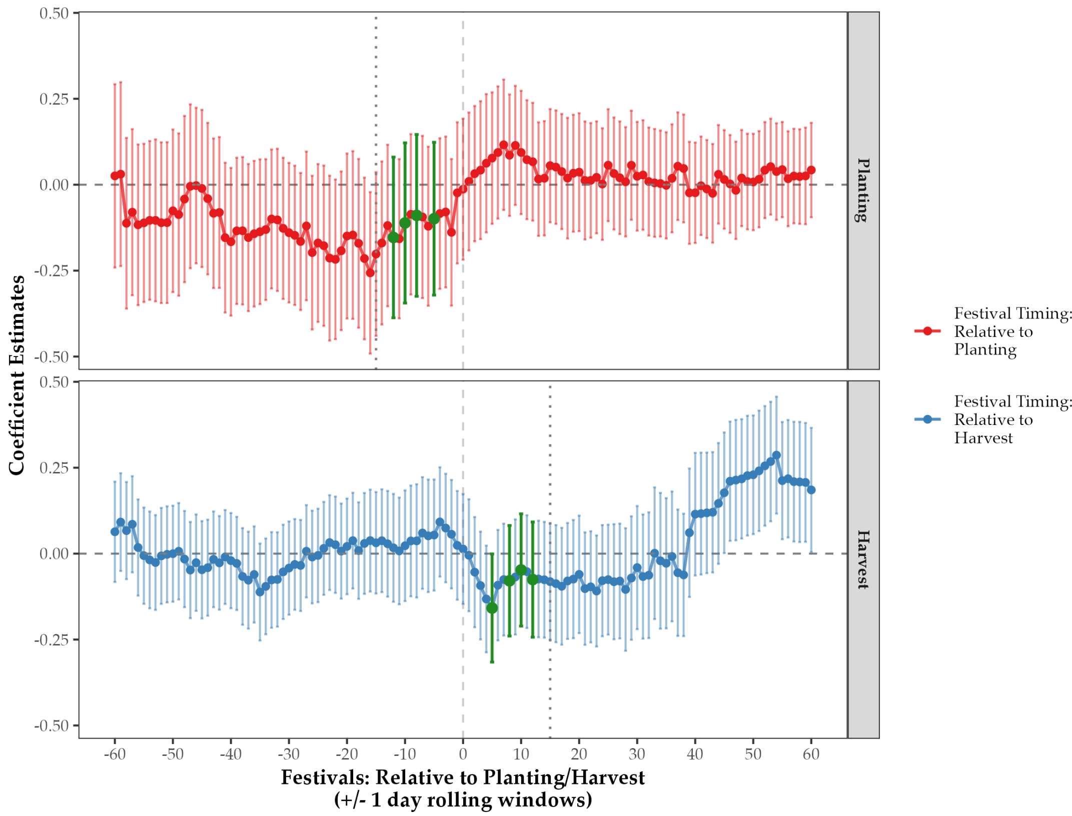
Patterns of effects are different for harvest-coinciding festivals in ways that are also revealing of mechanisms. Effects of harvest-coinciding festivals are largest for those occurring *after* harvest, suggesting that key mechanisms have to do with the *use* of harvest revenues – it may be that festivals happening soon after harvest lead to higher temptation spending of the harvest revenue, and thus less saving and investment in the future.

Campbell et al. (2024b) run versions of the analyses we conducted in Appendix Figure C4, perturbing the 30-day festival date windows in the exact directions in which we found that effects attenuate: perturbing planting date windows *after* the optimal planting date, and perturbing harvest date windows *before* the optimal harvest date. They show that coefficients attenuate when making these perturbations. This is precisely what we had also pointed out in Montero and Yang (2022) when discussing the results presented in Appendix Figure C4.

In response to Campbell et al. (2024b), we also provide an additional figure in this corrigendum (Figure R1) systematically examining a large set of 1-day perturbations. The figure shows coefficient estimates for planting-coinciding (top panel) and harvest-coinciding (bottom panel) festivals when perturbing the 30-day windows in 1-day shifts from 60 days prior to the optimal date to 60 days after the optimal date. The estimates of Campbell et al. (2024b) (3-, 5-, 7-, and 10-day perturbations) are colored in green. This figure provides more detail, but the conclusion is the same as from our Appendix Figure C4: effects are attenuated when perturbing planting date windows to *after* the optimal planting date, and perturbing harvest date windows to *before* the optimal harvest date.

In sum, the analyses of Campbell et al. (2024b) involving perturbing 30-day festival windows around optimal harvest and planting dates reveal patterns of effects that mirror

Figure R1: Impacts of Festival Timing Relative to Planting and Harvest



Notes: Data are from the 2010 Mexico Population Census for the *New Spain* region of Mexico. The figure presents the estimated β_i (top panel) and γ_i (bottom panel) coefficients and respective 95% confidence intervals from estimating equation (A1) for $i \in -60{,}60$ days. The outcome variable is *Log Household Income*. *Festival :* $i \pm 15$ *days from Planting* (top panel) is an indicator variable equal to 1 if the festival occurs $i \pm 15$ from the optimal planting date according to FAO GAEZ data (where negative values of i means that the festival occurs prior to planting); *Festival :* $i \pm 15$ *days from Harvest* (bottom panel) is an indicator variable equal to 1 if the festival occurs $i \pm 15$ from the optimal harvest date according to FAO GAEZ data (where negative values of i means that the festival occurs prior to harvest). The regressions control for the full-set of controls: *State Fixed Effects*, *Geography Controls*, *Colonial Controls*, *Festival-Week Fixed Effects*, and *Planting- & Harvest-Month Fixed Effects*.

those we highlighted in our original paper. These should be viewed as revealing potential underlying mechanisms behind the effects of coinciding festivals. It would be odd to present these as "robustness tests", particularly as they are versions of results that we already presented and discussed in the original paper.[6]

3.3.2. *Varying the Sample*

In this section, Campbell et al. (2024b) show how estimates vary when changing the subsample of municipalities in the analysis. They place the greatest emphasis on subsamples below and above median maize suitability. They find an interesting pattern of which we were not aware. It appears that the effect of coinciding festivals is a U-shape in maize suitability: close to zero for low and high maize suitability, and negative for intermediate levels. It appears that in places with the highest maize suitability, departures from optimal planting or harvest dates (and other types of interference from coinciding festivals) are less consequential for agricultural productivity. There may be more "wiggle room" to make suboptimal agricultural decisions in such high-productivity areas.

We find this to be an interesting dimension of treatment effect heterogeneity. It seems unusual, though, to describe this as a "robustness test". It seems more appropriate to view this analysis as an exploration of heterogeneity in the treatment effect.

3.3.3. *Robustness: Varying Controls*

In this section Campbell et al. (2024b) show results are robust to inclusion of "region" fixed effects. This seems redundant, since we already include state fixed effects in our regressions (which should be more detailed than "regions").

They also show results are robust to exclusion of certain selected controls. This is fine, but it does seem that the controls selected for exclusion are a bit arbitrary. This analysis also seems somewhat redundant given that in our main results tables we already systematically show how coefficient estimates vary across five different regression specifications with increasingly inclusive sets of control variables and fixed effects.

3.3.4. *Influential Analysis*

In this subsection, Campbell et al. (2024b) test robustness to exclusion of high-leverage observations on the basis of their dfbeta statistics. This analysis is also redundant: in the original paper (and revised here with corrected data) we conducted a systematic analysis

[6] In this subsection, Campbell et al. (2024b) also run regressions that combine shifts in festival date windows with inclusion of local saints and with analysis of impacts of other festivals. We discuss these types of analyses above in this corrigendum: on local saints, see Section 3.3, and on other festivals, see Section 2.2.4.

examining whether our key results are likely the result of outliers or high-leverage observations, following Young (2019). We simulate the data to examine the distribution of placebo treatment effects that would emerge with randomly assigned placebo coinciding festivals. In the original paper, the discussion is on p. 3202, and we discuss the revised results above in Section 2.1.1. Results of the Young (2019) randomization inference exercise are presented in Figure C2: the estimated t-statistic of our coinciding-festival coefficient is much more negative than the majority of placebo festival assignments, with a randomization inference p-value is 0.0036. Our results are unlikely to be driven by outliers or high-leverage observations.[7]

4. Conclusion

We thank Campbell et al. (2024b) for their efforts to enhance transparency in economics research. Such efforts provide important public goods for the profession. We are happy to engage with the authors in their efforts.

The most important outcome of their investigation of our paper, Montero and Yang (2022), is that they identified data-coding errors in our original analyses. We appreciate the opportunity to share revisions of all figures, tables, and empirical analyses in this corrigendum that use the corrected data. All key coefficient estimates are consistent in sign, remain statistically significant at conventional levels, and yield the same overall conclusions as the original paper.

We also address other critiques made by Campbell et al. (2024b). A number of their critiques are misplaced. The use of official Vatican-prescribed festival dates (rather than municipalities' actual celebrated dates) is intentional on our part, and is one of the underpinnings of our strategy for causal identification. Their analyses showing that coefficient estimates attenuate when making certain perturbations to planting- and harvest-festival date windows are versions of findings in our original paper that are revealing of underlying mechanisms; these should not be considered "robustness tests". Analyses showing differences in treatment effects for places with varying maize suitability reveal interesting patterns of treatment effect heterogeneity, and so are not properly described as "robustness tests".

A key conclusion of Campbell et al. (2024b) is that coefficient estimates are "robust at a 5% confidence level in 24% of the robustness checks we ran." This 24% figure is not reliable. Numerous analyses in Campbell et al. (2024b) are not properly considered "robustness tests". A number should be considered analyses of treatment effect hetero-

[7] Campbell et al. (2024b) include some other assorted regressions in this section. We note that the discussion in the text of columns 4 and 5 of their Table 10 is inconsistent with the statistics in the table, and that for the column 5 regression they refer to suitability for "wheat" when they probably mean "maize".

geneity. Others closely follow analyses we already conducted in the original paper whose role is to explore mechanisms. Some other analyses they present – such as regressions including municipalities with "local saints" – raise endogeneity concerns (and also are similar to robustness tests we already conducted in the original paper).

There are of course also more difficult questions about how one should systematically define the set of "robustness tests" to conduct in exercises such as Campbell et al. (2024b). The set of robustness tests to run can often seem *ad hoc*, and it can be inherently difficult to approach such decisions in a systematic way. We are sympathetic to such concerns. We defer such concerns to the broader discussion of the authors' umbrella paper – Campbell et al. (2024a) – critiquing the full set of 17 *American Economic Review* papers.

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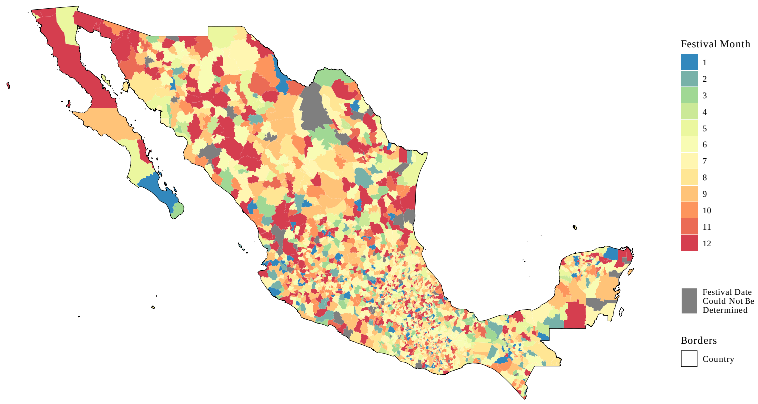
5. Main Paper Tables & Figures

Figure 1: Saint Day Festival Months - All of Mexico

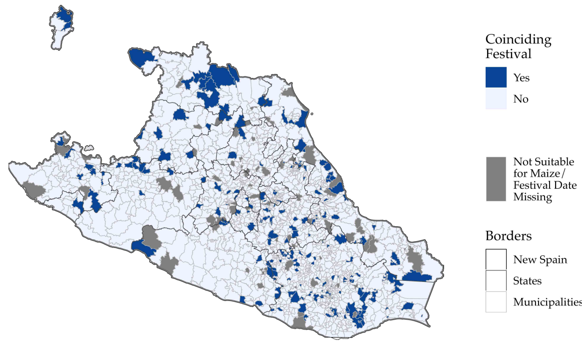


Notes: The map presents the month that each municipality in Mexico celebrates its respective Catholic patron saint day festival. Municipalities where we were unable to determine the festival date are shaded in dark grey. See Appendix A.1 for more information on the construction of the festival date dataset.

Figure 2: Administrative Borders and New Spain Region of Mexico

Notes: Administrative borders of Mexico in shades of gray: country, state, and municipality borders. In black: border of the New Spain region in 1786 (Gerhard, 1993).

Figure 3: Agriculturally-Coinciding Festivals - *New Spain* Region of Mexico



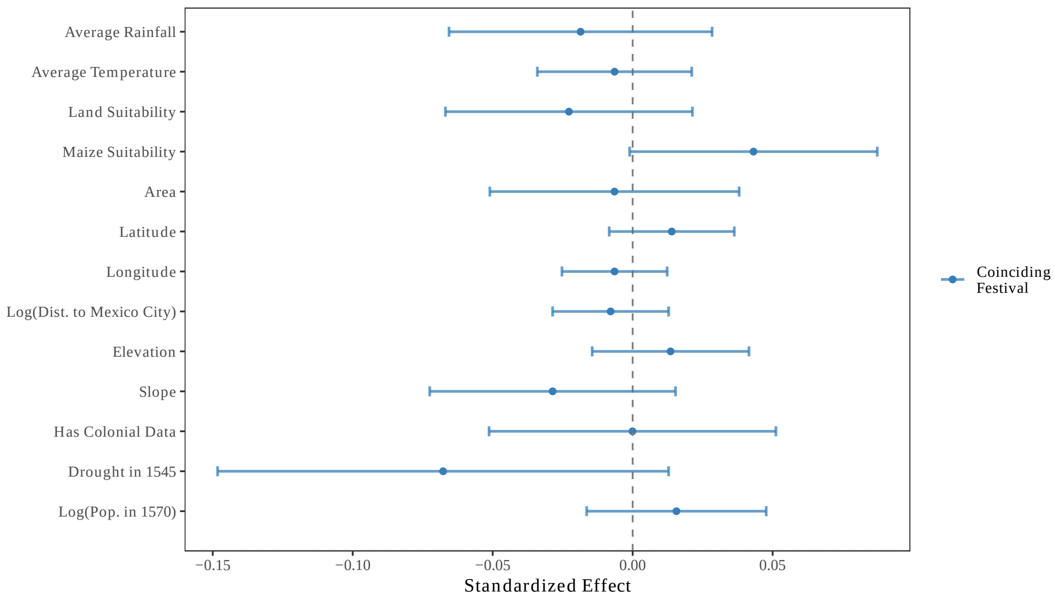
Notes: Coinciding Festival is equal to "Yes" if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data, and "No" otherwise for each municipality in the New Spain region of Mexico. Municipalities where we were unable to determine the festival date or are unsuitable for maize are shaded in dark grey.

Table 1: Relationship Between Festival, Planting, and Harvest Months

	Dependent Variable: *Festival Date*			
	(1)	(2)	(3)	(4)
Maize Planting or Harvest Month	−0.022	−0.016		
	(0.017)	(0.017)		
Maize Planting Month			−0.021	−0.011
			(0.020)	(0.022)
Maize Harvest Month			−0.022	−0.020
			(0.027)	(0.027)
Calendar Week Fixed Effects	Y	N	Y	N
Week by State Fixed Effects	N	Y	N	Y
Observations	583,038	583,038	583,038	583,038
Clusters	1,593	1,593	1,593	1,593
Adjusted R2	0.004	0.004	0.004	0.004
Mean Dep. Var.	0.273	0.273	0.273	0.273

Notes: Observations are at the municipality-calendar date level for municipalities in the *New Spain* region of Mexico for which we have festival data. Standard errors clustered at the municipality level are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. For ease of interpretation, we multiply all regression coefficients by 100. *Maize Planting or Harvest Month* is an indicator variable equal to 1 if a date falls within 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Maize Planting Month* is an indicator variable equal to 1 if a date falls within 0 to 30 days prior to the optimal maize planting date for a municipality using FAO GAEZ data. *Maize Harvest Month* is an indicator variable equal to 1 if a date falls within 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Calendar Week Fixed Effects* are indicator variables for each separate 7-day period in the calendar year (52 fixed effects). *Week by State Fixed Effects* are interaction terms between calendar week and Mexican state fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4: Municipality Characteristics and Coinciding Festivals



Notes: Data are from the 2010 Mexico Population Census for *New Spain* region of Mexico. Colonial drought and population density data are from Sellers and Alix-Garcia (2018). The figure presents the estimated standardized coefficients and respective 95% confidence intervals from estimating equation (1) on various municipality characteristics (denoted on the y-axis), conditional on state fixed effects, planting-month and harvest-month fixed effects, and festival-week fixed effects. *Coinciding Festival* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either within 0-30 days prior to the optimal maize planting date or 0-30 days after the optimal maize harvest date for a municipality using FAO GAEZ data, and 0 otherwise. Note that we do not have colonial characteristics for all observations in our sample; therefore, we also show results for *Has Colonial Characteristics*, an indicator equal to 1 if a municipality is not missing colonial characteristics.

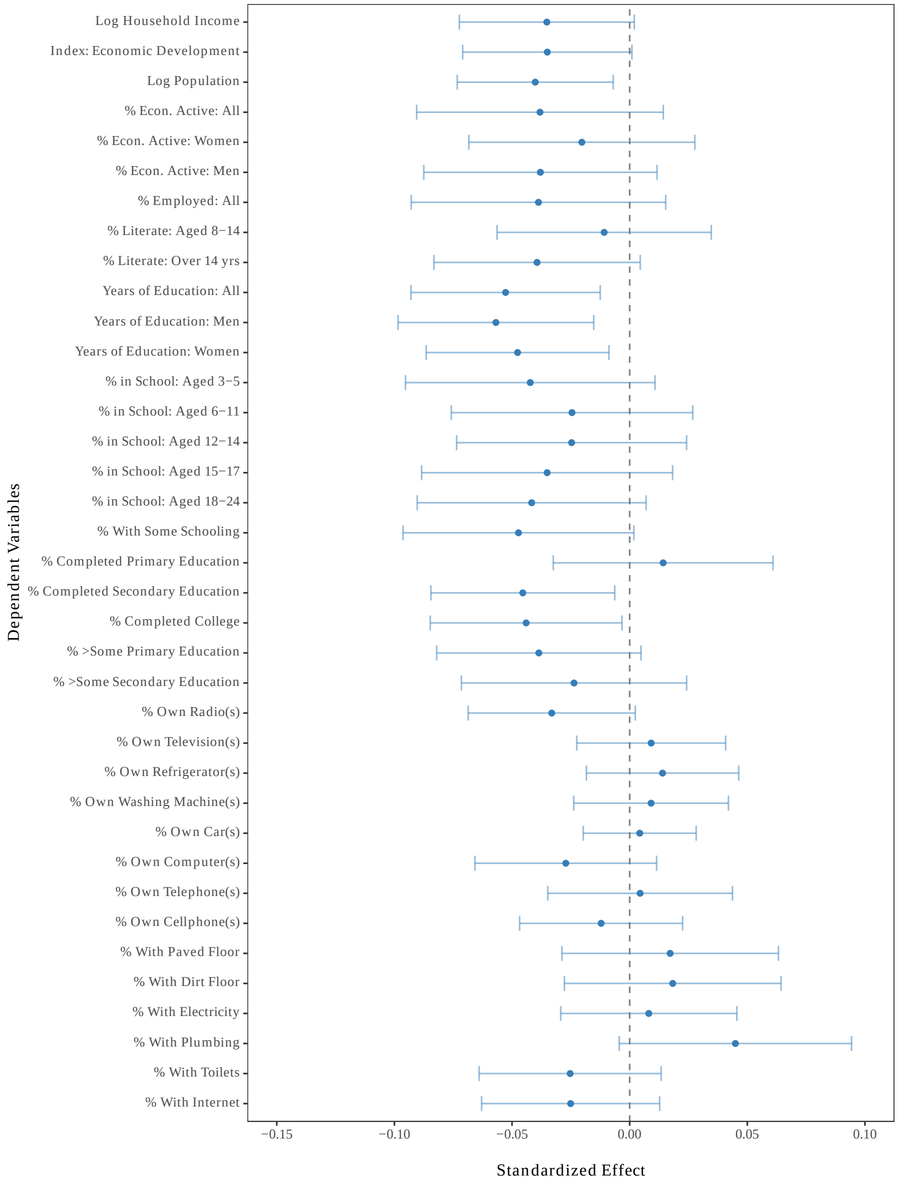
Table 2: Impact of Agriculturally-Coinciding Festivals on Development Outcomes

	Dependent Variable: Panel A: *Log HH Income*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.208	−0.134	−0.178	−0.182	−0.132
	(0.091)	(0.077)	(0.066)	(0.066)	(0.071)
	[0.097]	[0.066]	[0.058]	[0.056]	[0.058]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,593	1,593	1,593	1,593	1,593
Adjusted R2	0.002	0.344	0.535	0.541	0.562
Mean Dep. Var.	2.792	2.792	2.792	2.792	2.792
SD Dep. Var.	1.228	1.228	1.228	1.228	1.228

	Dependent Variable: Panel B: *Index of Economic Development*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.581	−0.284	−0.472	−0.496	−0.438
	(0.306)	(0.254)	(0.220)	(0.218)	(0.230)
	[0.343]	[0.239]	[0.193]	[0.190]	[0.167]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,593	1,593	1,593	1,593	1,593
Adjusted R2	0.002	0.347	0.565	0.571	0.597
Mean Dep. Var.	-0.591	-0.591	-0.591	-0.591	-0.591
SD Dep. Var.	4.039	4.039	4.039	4.039	4.039

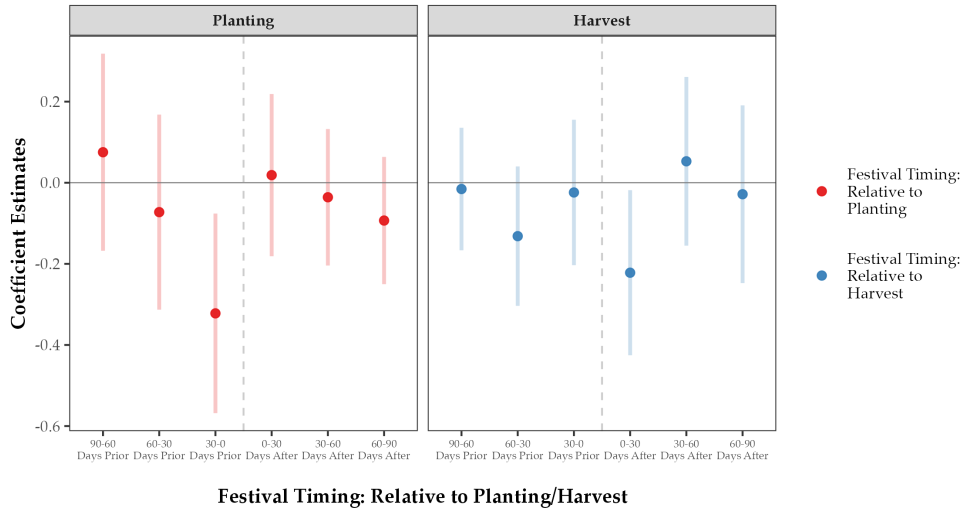
Notes: Data are from the 2010 Mexico Population Census. Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Index of Economic Development* is the first principal component index of a number of development outcomes in the census for a municipality (see Appendix A). *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* are fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar week of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 5: Impact of Agriculturally-Coinciding Festivals on Development Outcomes: Estimates for *Economic Development Index* Components



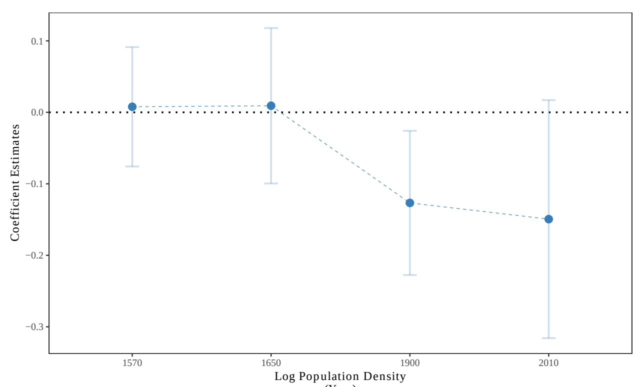
Notes: Data are from the 2010 Mexico Population Census for *New Spain* region of Mexico. The figure presents the estimated standardized coefficients and respective 95% confidence intervals from estimating equation (1) on the sub-components of the *Index of Economic Development*. The dependent variables are denoted on the y-axis. We first show the estimates for log household income, then for the *Index of Economic Development*, followed by estimates for each of the individual sub-components of the index (note, log household income is also one of the index components). See Data Appendix for more information. The independent variable is *Festival Coincides with Maize Planting or Harvest*: an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. The regressions control for the full set of controls: *State Fixed Effects*, *Geography Controls*, *Colonial Controls*, *Festival-Week Fixed Effects*, and *Planting- & Harvest-Month Fixed Effects*.

Figure 6: Impacts of Festival Coincidence with Other Months Relative to Planting and Harvest



Notes: Data are from the 2010 Mexico Population Census for the *New Spain* region of Mexico. The figure presents the estimated β_i and γ_i coefficients and respective 95% confidence intervals from estimating equation (3). The outcome variable is *Log Household Income*. *Festival Timing: Relative to Planting/Harvest* is defined as the number of months before/after a municipality celebrates its festival relative to planting (top panel) and harvest (bottom panel) according to FAO GAEZ data. The regressions control for state fixed effects, geography controls and colonial controls.

Figure 7: Impact of Agriculturally-Coinciding Festivals on Population



Notes: Data on log population density for 1570, 1650, and 1900 is from Sellers and Alix-Garcia (2018). Data on population density for 2010 is from the 2010 Mexico Population Census. The figure presents the estimated coefficients and respective 95% confidence intervals from estimating equation (1) for various measures of log population density across time. Observations throughout are municipalities in the *New Spain* region of Mexico that have historical population density (1,395 municipalities in each year). The independent variable is *Festival Coincides with Maize Planting or Harvest*, an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to optimal maize planting month or 0 to 30 days after the optimal maize harvest month for that municipality using FAO GAEZ data. The regressions control for the full set of controls: *State Fixed Effects*, *Geography Controls*, *Colonial Controls*, *Festival-Week Fixed Effects*, and *Planting- & Harvest-Month Fixed Effects*.

Table 3: Impact of Agriculturally-Coinciding Festivals on Agricultural Productivity and Structural Transformation

	Dependent Variables:			
	Maize Yield	*% in Agriculture*	*% in Manufacturing*	*% in Services*
	(1)	(2)	(3)	(4)
Festival Coincides with Maize Planting or Harvest	−0.421	0.024	−0.004	−0.019
	(0.216)	(0.015)	(0.008)	(0.013)
	[0.194]	[0.013]	[0.008]	[0.011]
State Fixed Effects	Y	Y	Y	Y
Geography Controls	Y	Y	Y	Y
Colonial Controls	Y	Y	Y	Y
Planting-Month Fixed Effects	Y	Y	Y	Y
Harvest-Month Fixed Effects	Y	Y	Y	Y
Festival-Week Fixed Effects	Y	Y	Y	Y
Observations	1,580	1,593	1,593	1,593
Adjusted R2	0.470	0.485	0.233	0.435
Mean Dep. Var.	6.128	0.414	0.114	0.465
SD Dep. Var.	4.140	0.237	0.095	0.198

Notes: Maize yield data are from the *Servicio de Información Agroalimentaria y Pesquera* (SIAP) for 2010, and other outcomes from the 2010 Population Census. Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Maize Yield* is the mean maize revenue yield in thousand of pesos per hectare for a municipality in 2010. *% in Agriculture* is the share of workers in a municipality who work in agriculture. *% in Manufacturing* is the share of workers in a municipality who work in manufacturing. *% in Services* is the share of workers in a municipality who work in the service industry. *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar week of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Impact of Agriculturally-Coinciding Festivals on Religiosity, Social Capital, and Inequality

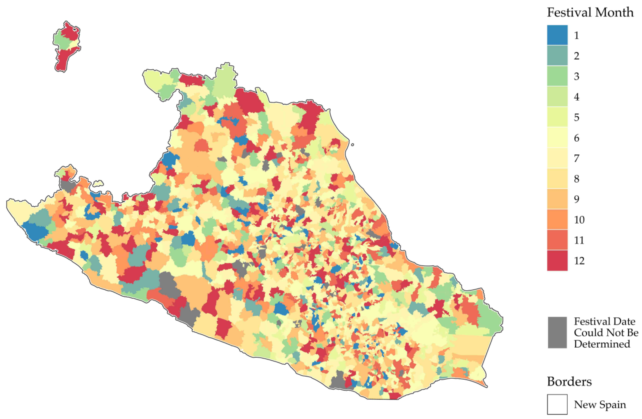
	Dependent Variables:		
	Religiosity Index	*Group Membership Index*	*IQR of Earned Incomes*
	(1)	(2)	(3)
Festival Coincides with Maize Planting or Harvest	0.315	0.240	−0.242
	(0.162)	(0.135)	(0.106)
	[0.155]	[0.125]	[0.095]
State Fixed Effects	Y	Y	Y
Geography Controls	Y	Y	Y
Colonial Controls	Y	Y	Y
Planting-Month Fixed Effects	Y	Y	Y
Harvest-Month Fixed Effects	Y	Y	Y
Festival-Week Fixed Effects	Y	Y	Y
Observations	4,796	4,818	1,593
Clusters	131	131	1593
Adjusted R2	0.129	0.050	0.555
Mean Dep. Var.	-0.187	0.063	2.168
SD Dep. Var.	1.266	1.306	1.833

Notes: Data are from the Americas Barometer (LAPOP) data (religiosity and social capital) and from the 2010 Mexico Population Censuses from IPUMS (inequality). Observations are individuals in municipalities in the *New Spain* region of Mexico. Standard errors clustered at the municipality level are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Religiosity Index* is the first principal component of the following variables: *Importance of Religion*, *Church Attendance*, and *Religious Group Attendance*. *Importance of Religion* is a 1-4 categorical variable that measures how important religion is to a respondent, ranging from 1="Not Important at All" to 4="Very Important". *Church Attendance* is a 1-5 categorical variable that measures how frequently an individual goes to church, ranging from 1="Never" to 5="More than Once a Week". *Religious Group Attendance* is a 1-4 categorical variable that measures how frequently an individual participates in religious group meetings, ranging from 1="Never" to 4="Once a Week". *Group Membership Index* is the first principal component for the frequency with which a respondent participates in the following group meetings: community improvement, parental associations, municipal meetings, or political associations. *IQR of Earned Incomes* measures the inter-quartile range of individuals total income from their labor (from wages, a business, or a farm) in the previous month for individuals residing in a given municipality. *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. All regressions controls for respondent age, age squared, gender, and include survey-wave fixed effects. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar week of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B. Additional Maps

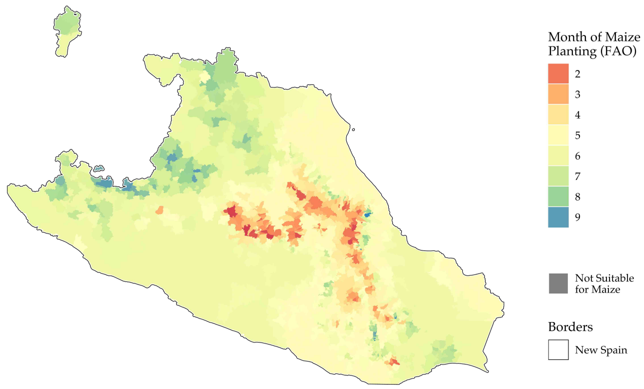
B.1. Additional Maps – New Spain

Figure B1: Saint Day Festival Months: *New Spain* Region of Mexico



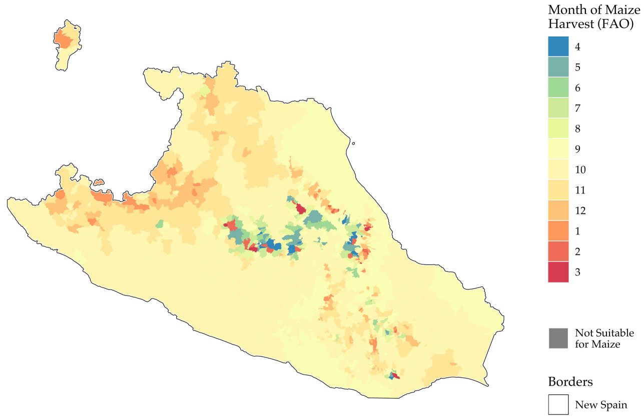
Notes: The map presents the month that each municipality in the New Spain region of Mexico celebrates its respective Catholic patron saint day festival. Municipalities where we were unable to determine the festival date are shaded in dark grey. Borders of New Spain region of colonial Mexico are as defined by Gerhard (1993). See Appendix A.1 for more information on the construction of the festival date dataset.

Figure B2: Maize Planting Month (FAO data):
New Spain Region of Mexico



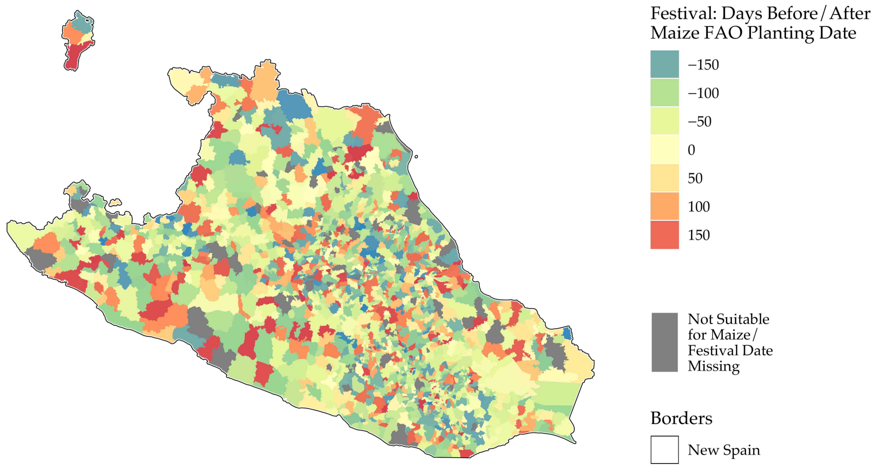
Notes: Optimal maize planting month according to FAO GAEZ data in the New Spain region of Mexico.

Figure B3: Maize Harvest Month (FAO data):
New Spain Region of Mexico



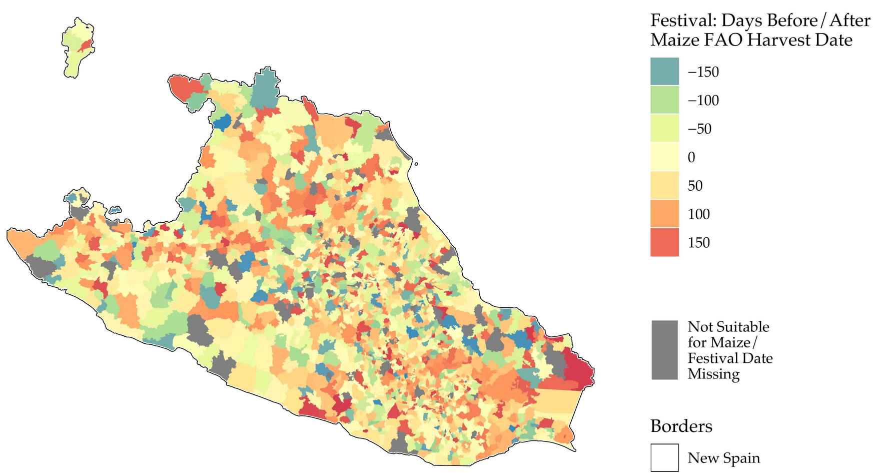
Notes: Optimal maize harvest month according to FAO GAEZ data in the New Spain region of Mexico.

Figure B4: Days Between Festival and Optimal Planting Date:
New Spain Region of Mexico



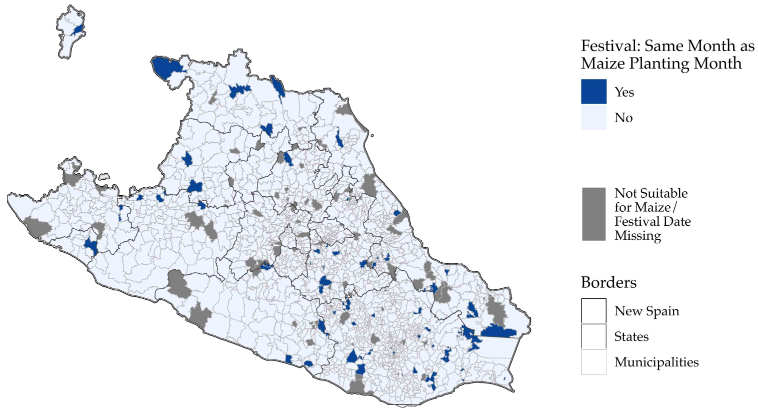
Notes: Difference (in days) between the patron saint day festival date and the optimal maize planting date (from FAO GAEZ data) for each municipality in the New Spain region of Mexico. (Negative values correspond to festivals that occur before planting; positive values correspond to festivals that occur after planting.) Municipalities where we were unable to determine the festival date are shaded in dark grey.

Figure B5: Days Between Festival and Optimal Harvest Date:
New Spain Region of Mexico



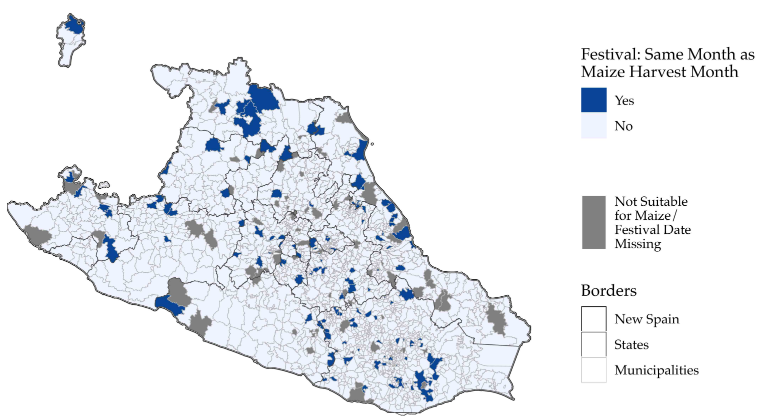
Notes: Difference (in days) between the patron saint day festival date and the optimal maize harvest date (from FAO GAEZ data) for each municipality in the New Spain region of Mexico. (Negative values correspond to festivals that occur before harvest; positive values correspond to festivals that occur after harvest.) Municipalities where we were unable to determine the festival date are shaded in dark grey.

Figure B6: Coincidence of Festivals and Optimal Planting Month:
New Spain Region of Mexico



Notes: The map presents whether the month that each municipality in the *New Spain* region of Mexico celebrates its respective patron saint day festival falls 0-30 days prior to the optimal maize planting date according to FAO GAEZ data. Municipalities where we were unable to determine the festival date are shaded in dark grey.

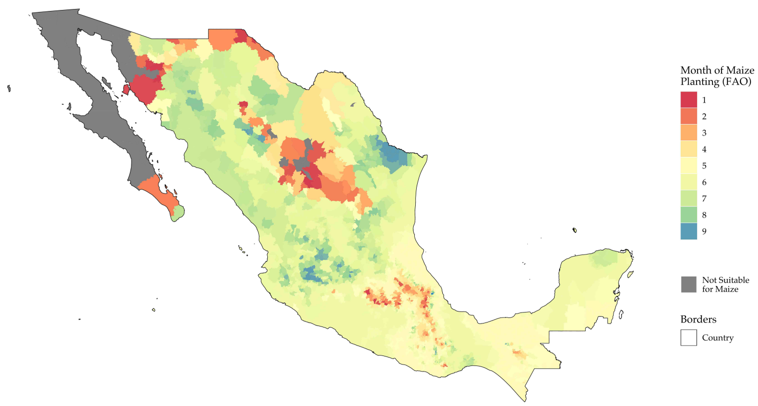
Figure B7: Coincidence of Festivals and Optimal Harvest Month:
New Spain Region of Mexico



Notes: The map presents whether the month that each municipality in the *New Spain* region of Mexico celebrates its respective patron saint day festival falls 0-30 days after the optimal maize harvest date according to FAO GAEZ data. Municipalities where we were unable to determine the festival date are shaded in dark grey.

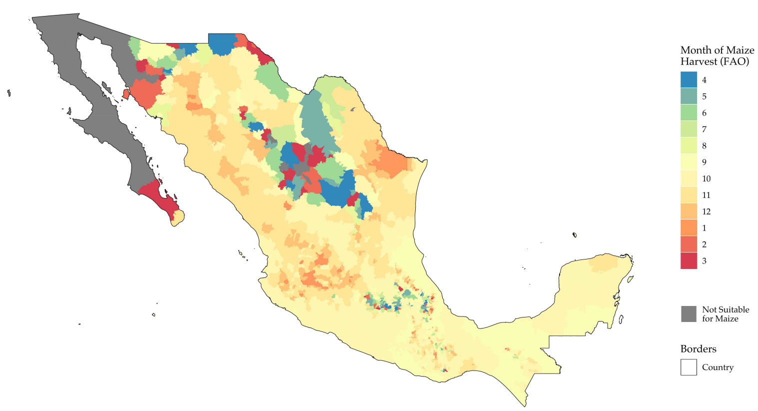
B.2. Additional Maps – All of Mexico

Figure B8: Maize Planting Date (FAO data)



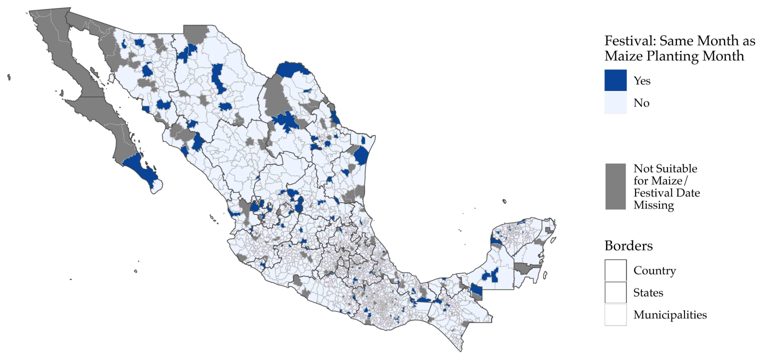
Notes: Optimal maize planting month according to FAO GAEZ data for each municipality in Mexico.

Figure B9: Maize Harvest Date (FAO data)



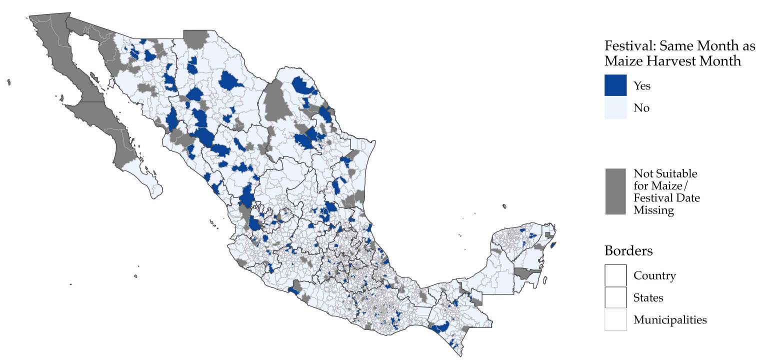
Notes: Optimal maize harvest month according to FAO GAEZ data for each municipality in Mexico.

Figure B10: Coincidence of Festivals and Optimal Planting Month



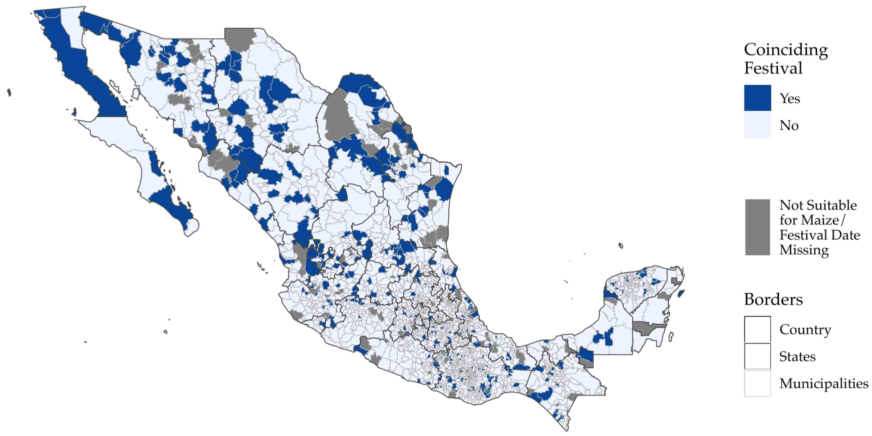
Notes: The map presents whether the date that each municipality in Mexico celebrates its respective patron saint day festival falls 0-30 days prior to the optimal maize planting date according to FAO GAEZ data. Municipalities where we were unable to determine the festival date or are unsuitable for maize are shaded in dark grey.

Figure B11: Coincidence of Festivals and Optimal Harvest Month



Notes: The map presents whether the date that each municipality in Mexico celebrates its respective patron saint day festival falls 0-30 days after the optimal maize harvest date according to FAO GAEZ data. Municipalities where we were unable to determine the festival date or are unsuitable for maize are shaded in dark grey.

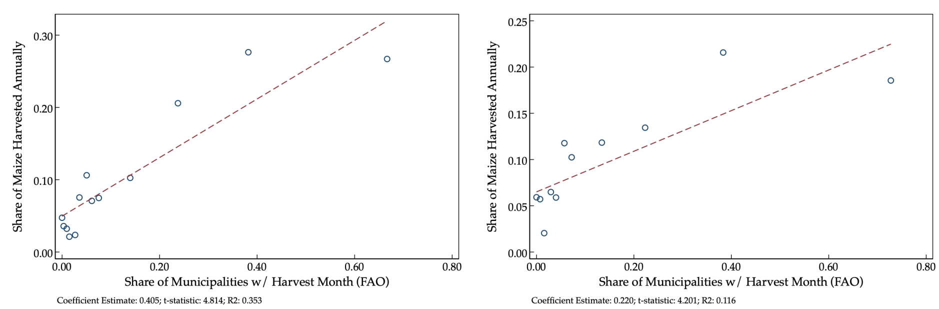
Figure B12: Agriculturally-Coinciding Festivals



Notes: Coinciding Festival is equal to "Yes" if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data and "No" otherwise for each municipality in Mexico. Municipalities where we were unable to determine the festival date or are unsuitable for maize are shaded in dark grey.

Appendix C. Additional Figures

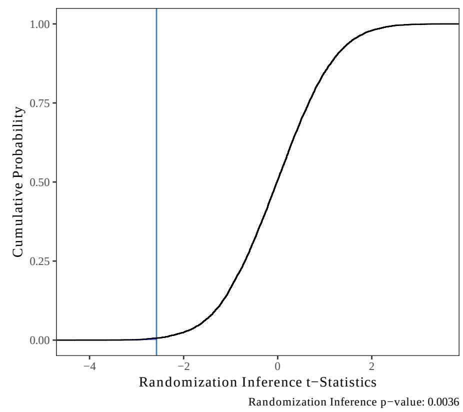
Figure C1: Validating Crop Calendar Data:
Relationship Between Predicted Maize Harvest Timing and Actual Maize Harvest



a. *New Spain* Region　　　b. All of Mexico

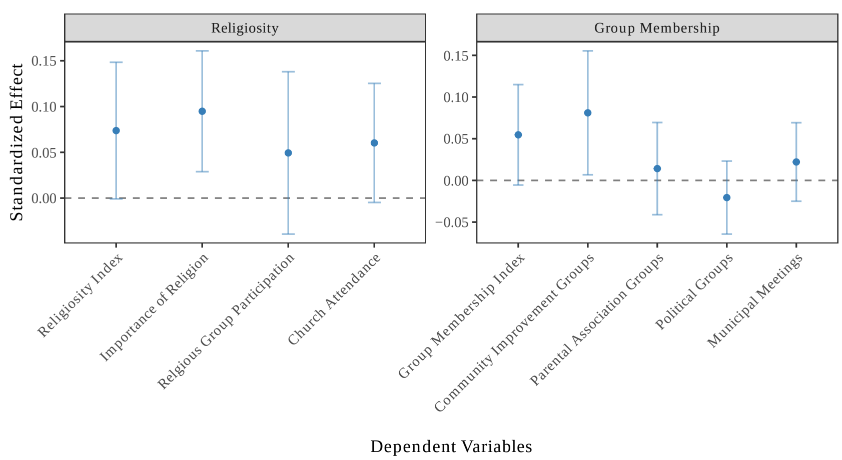
Notes: The figure presents binscatters between the share of a state's total maize harvest that occurs on a given month and the share of municipalities in a state that have their maize harvest on a given month according to the FAO GAEZ data. The unit of observation is a state-month pair. State harvest Data are from the *Servicio de Información Agroalimentaria y Pesquera (SIAP)* for 2015. The bottom-right of each figure presents the estimated bivariate coefficient, t-statistic, and R^2. Standard errors are clustered at the state level.

Figure C2: Randomization Inference Exercise – Placebo Festivals



Notes: The figure presents the cumulative distribution function for the estimated t-statistics for the randomization inference exercise. Specifically, we conduct 10,000 simulations where we randomly assign whether or not a festival coincides with planting or harvest for each municipality and estimate our main specification, and then plot the cumulative distribution function for the estimated t-statistics. The dependent variable is *Log Household Income*. All regressions include state fixed effects, *Geography Controls*, *Colonial Controls*, *Festival-Week Fixed Effects*, and *Planting- & Harvest-Month Fixed Effects*. Observations are municipalities in the *New Spain* region of Mexico. Additionally, the figure presents the estimated t-statistic for our sample, and reports the randomization inference p-value on the bottom right of the figure.

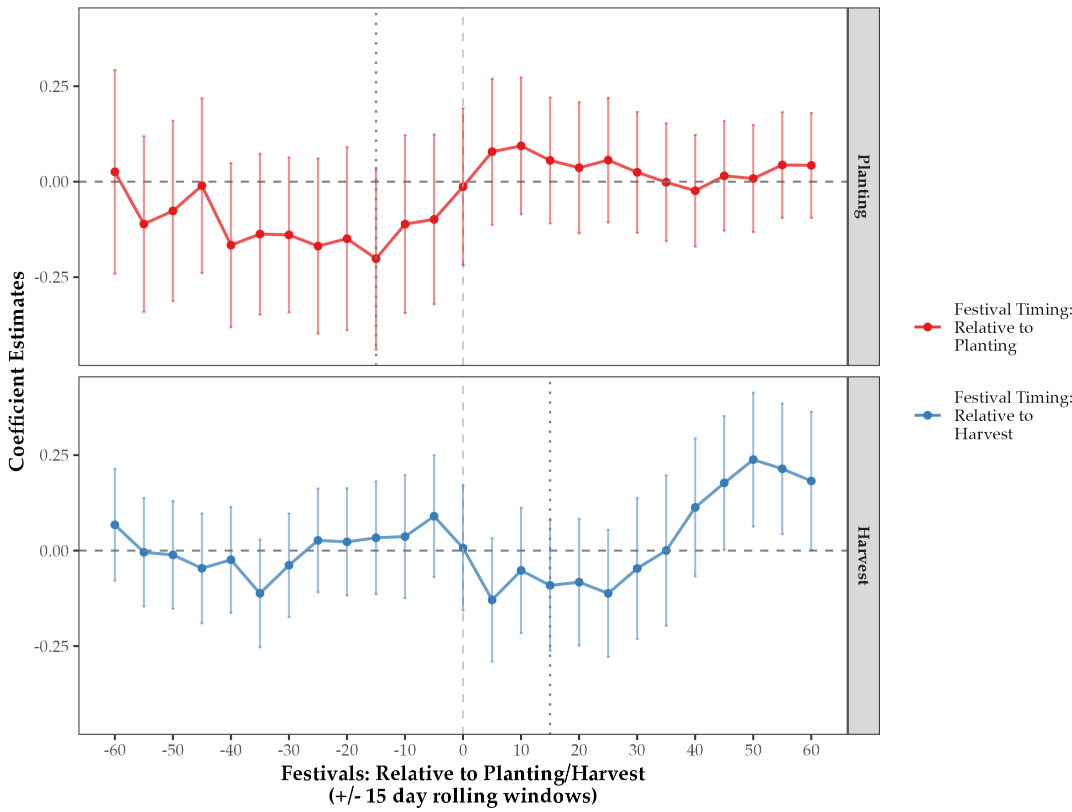
Figure C3: Impact of Coinciding Festivals on Religiosity and Social Capital: Estimates for *Religiosity Index* and *Group Membership Index* Components



Notes: Data are from the Americas Barometer (LAPOP) data for *New Spain* region of Mexico. The figure presents the estimated coefficients and respective 95% confidence intervals from estimating equation (1) on the sub-components of the *Religiosity Index* and the *Group Membership Index*. The dependent variables are denoted on the x-axis. We first show the estimates for each index, followed by estimates for each of the individual sub-components of the index. (See Data Appendix for more information.) The independent variable is *Festival Coincides with Maize Planting or Harvest*: an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. The regressions control for respondent age, age squared, gender, and for the following set of controls: *Survey-Wave Fixed Effects*, *State Fixed Effects*, *Geography Controls*, *Colonial Controls*, *Festival-Week Fixed Effects*, and *Planting- & Harvest-Month Fixed Effects*.

Sensitivity to 30-Day Window Used to Define Coinciding with Planting or Harvest Months

Figure C4: Impacts of Festival Timing Relative to Planting and Harvest Dates



Notes: Data are from the 2010 Mexico Population Census for the *New Spain* region of Mexico. The figure presents the estimated β_i (top panel) and γ_i (bottom panel) coefficients and respective 95% confidence intervals from estimating equation (A1) for $i \in -60{,}60$ days. The outcome variable is *Log Household Income*. *Festival :* $i \pm 15$ *days from Planting* (top panel) is an indicator variable equal to 1 if the festival occurs $i \pm 15$ from the optimal planting date according to FAO GAEZ data (where negative values of i means that the festival occurs prior to planting); *Festival :* $i \pm 15$ *days from Harvest* (bottom panel) is an indicator variable equal to 1 if the festival occurs $i \pm 15$ from the optimal harvest date according to FAO GAEZ data (where negative values of i means that the festival occurs prior to harvest). The regressions control for the full-set of controls: *State Fixed Effects*, *Geography Controls*, *Colonial Controls*, *Festival-Week Fixed Effects*, and *Planting- & Harvest-Month Fixed Effects*.

Appendix D. Additional Tables

Table D1: Relationship Between Festival, Planting, and Harvest Months: All of Mexico

	Dependent Variable: *Festival Date*			
	(1)	(2)	(3)	(4)
Maize Planting or Harvest Month	0.009	0.008		
	(0.015)	(0.016)		
Maize Planting Month			0.015	0.019
			(0.019)	(0.021)
Maize Harvest Month			0.003	−0.003
			(0.023)	(0.024)
Calendar Date Fixed Effects	Y	N	Y	N
Date by State Fixed Effects	N	Y	N	Y
Observations	833,382	833,382	833,382	833,382
Clusters	2,277	2,277	2,277	2,277
Mean Dep. Var.	0.273	0.273	0.273	0.273

Notes: Observations are at the municipality-month level for municipalities in Mexico for which we have festival data. Standard errors clustered at the municipality level are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. For ease of interpretation, all coefficients are multiplied by 100. *Maize Planting or Harvest Month* is an indicator variable equal to 1 if a date falls within 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data.*Maize Planting Month* is an indicator variable equal to 1 if a date falls within 0 to 30 days prior to the optimal maize planting date for a municipality using FAO GAEZ data. *Maize Harvest Month* is an indicator variable equal to 1 if a date falls within 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Calendar Week Fixed Effects* are indicator variables equal to 1 if the festival for a municipality occurs in a given 7-day period (52 fixed effects). *Week by State Fixed Effects* are interaction terms between calendar week and Mexican state fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D2: Municipality Characteristics and Coinciding Festivals

	Non-Coinciding Festival			*Coinciding Festival*			Regression Estimates: *Coinciding Festival*		
	Obs.	Mean	SE	Obs.	Mean	SE	Coef.	Robust SE	Conley SE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Geographic Characteristics:									
Precipitation	1404	95.18	(1.22)	189	97.29	(3.58)	-2.91	(3.46)	[3.08]
Temperature	1404	19.05	(0.11)	189	19.63	(0.27)	-0.09	(0.17)	[0.17]
Land Suitability	1404	86.48	(0.34)	189	84.69	(1.25)	-0.78	(0.94)	[0.99]
Maize Suitability	1404	34.47	(0.58)	189	35.04	(1.62)	2.99	(1.54)	[1.46]
Area	1404	328.59	(13.22)	189	338.24	(39.81)	-3.49	(35.95)	[35.75]
Longitude	1404	-98.33	(0.05)	189	-98.22	(0.15)	-0.05	(0.06)	[0.06]
Latitude	1404	18.65	(0.04)	189	18.77	(0.12)	0.06	(0.05)	[0.05]
Log(Dist. to Mexico City)	1404	5.42	(0.02)	189	5.52	(0.05)	-0.02	(0.02)	[0.02]
Slope	1404	10.35	(0.17)	189	9.54	(0.46)	-0.55	(0.45)	[0.44]
Elevation	1404	1572.05	(20.95)	189	1468.80	(57.04)	36.65	(35.57)	[33.78]
Colonial Characteristics:									
Has Colonial Characteristics (%)	1404	86.04	(0.93)	189	83.07	(2.74)	0.38	(2.86)	[2.67]
Drought in 1545 (%)	1208	99.67	(0.17)	157	98.09	(1.10)	-1.48	(0.88)	[1.00]
Log(Pop. Density in 1570)	1208	0.53	(0.03)	157	0.43	(0.08)	0.06	(0.06)	[0.04]

Notes: Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Coinciding Festival* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either within 0-30 days prior to the optimal maize planting date or 0-30 days after the optimal maize harvest date for a municipality using FAO GAEZ data, and 0 otherwise. The value displayed for regression estimates is the coefficient estimate for *Coinciding Festival*, conditional on state fixed effects, planting-month and harvest-month fixed effects, and festival week fixed effects. Robust standard errors are presented in parentheses. See Data Appendix for more information on variables. Note that we do not have colonial characteristics for all observations in our sample; therefore, we also show results for *Has Colonial Characteristics*, an indicator equal to 1 if a municipality is not missing colonial characteristics. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D3: Municipality Characteristics and Coinciding Festivals: All of Mexico

	Non-Coinciding Festival			*Coinciding Festival*			Regression Estimates: *Coinciding Festival*		
	Obs.	Mean	SE	Obs.	Mean	SE	Coef.	Robust SE	Conley SE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Geographic Characteristics:									
Precipitation	1945	91.15	(1.12)	332	85.12	(2.82)	-4.08	(2.32)	[2.14]
Temperature	1945	19.64	(0.09)	332	20.12	(0.20)	-0.22	(0.15)	[0.14]
Land Suitability	1945	78.48	(0.59)	332	70.65	(1.78)	0	(0.77)	[0.73]
Maize Suitability	1945	31.39	(0.47)	332	29.53	(1.12)	1.45	(0.98)	[1.01]
Area	1945	824.83	(47.68)	332	1585.07	(255.60)	226.23	(158.42)	[153.73]
Longitude	1945	-98.56	(0.09)	332	-99.51	(0.29)	-0.06	(0.05)	[0.05]
Latitude	1945	19.77	(0.07)	332	21.01	(0.23)	0.10	(0.05)	[0.05]
Log(Dist. to Mexico City)	1945	5.80	(0.02)	332	6.10	(0.05)	-0.01	(0.02)	[0.01]
Slope	1945	9.25	(0.15)	332	8.28	(0.33)	-0.44	(0.32)	[0.32]
Elevation	1945	1405.12	(18.89)	332	1244.18	(45.00)	48.32	(29.67)	[27.24]
Colonial Characteristics:									
Has Colonial Characteristics (%)	1945	73.32	(1.00)	332	66.87	(2.59)	-0.70	(1.83)	[1.82]
Drought in 1545 (%)	1426	97.48	(0.42)	222	96.40	(1.25)	0.15	(1.33)	[1.48]
Log(Pop. Density in 1570)	1426	0.27	(0.03)	222	-0.07	(0.09)	0.02	(0.05)	[0.04]

Notes: Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Coinciding Festival* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either within 0-30 days prior to the optimal maize planting date or 0-30 days after the optimal maize harvest date for a municipality using FAO GAEZ data, and 0 otherwise. The value displayed for regressions estimates is the coefficient estimate for *Coinciding Festival*, conditional on state fixed effects, planting-month and harvest-month fixed effects, and festival week fixed effects. Robust standard errors are presented in parentheses. Robust standard errors are presented in parentheses. See Data Appendix for more information on variables. *Festival: Coincides* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either within 0-30 days prior to the optimal maize planting date or 0-30 days after the optimal maize harvest date for a municipality using FAO GAEZ data, and 0 otherwise. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D4: Development Outcomes and Planting- and Harvest-Coinciding Festivals

	Dependent Variable: Panel A: *Log HH Income*				
	(1)	(2)	(3)	(4)	(5)
Festival: 0-30 Days Prior to Maize Planting	−0.317	−0.116	−0.220	−0.242	−0.201
	(0.137)	(0.118)	(0.104)	(0.102)	(0.121)
	[0.142]	[0.099]	[0.104]	[0.103]	[0.113]
Festival: 0-30 Days After Maize Harvest	−0.141	−0.144	−0.152	−0.146	−0.095
	(0.115)	(0.095)	(0.082)	(0.082)	(0.087)
	[0.122]	[0.090]	[0.073]	[0.074]	[0.074]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,593	1,593	1,593	1,593	1,593
Adjusted R2	0.002	0.344	0.535	0.541	0.562
Mean Dep. Var.	2.792	2.792	2.792	2.792	2.792
SD Dep. Var.	1.228	1.228	1.228	1.228	1.228
P-Value: Difference	0.309	0.850	0.602	0.455	0.477

	Dependent Variable: Panel B: *Index of Economic Development*				
	(1)	(2)	(3)	(4)	(5)
Festival: 0-30 Days Prior to Maize Planting	−0.759	−0.015	−0.343	−0.426	−0.489
	(0.459)	(0.391)	(0.356)	(0.353)	(0.389)
	[0.500]	[0.348]	[0.359]	[0.358]	[0.373]
Festival: 0-30 Days After Maize Harvest	−0.471	−0.448	−0.550	−0.538	−0.411
	(0.388)	(0.315)	(0.270)	(0.268)	(0.283)
	[0.439]	[0.341]	[0.277]	[0.282]	[0.244]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,593	1,593	1,593	1,593	1,593
Adjusted R2	0.001	0.347	0.565	0.570	0.596
Mean Dep. Var.	-0.591	-0.591	-0.591	-0.591	-0.591
SD Dep. Var.	4.039	4.039	4.039	4.039	4.039
P-Value: Difference	0.620	0.372	0.635	0.795	0.870

Notes: Data is from the 2010 Mexico Population Census. Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Index of Economic Development* is the first principal component index for a number of development outcomes in the census for a municipality (see Data Appendix). *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar date of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D5: Impact of Agriculturally-Coinciding Festivals on Development Outcomes: All of Mexico

	Dependent Variable: Panel A: *Log HH Income*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	0.023	−0.064	−0.103	−0.107	−0.077
	(0.072)	(0.059)	(0.051)	(0.051)	(0.055)
	[0.085]	[0.054]	[0.048]	[0.047]	[0.047]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	2,277	2,277	2,277	2,277	2,277
Adjusted R2	-0.000	0.351	0.519	0.523	0.533
Mean Dep. Var.	2.932	2.932	2.932	2.932	2.932
SD Dep. Var.	1.214	1.214	1.214	1.214	1.214

	Dependent Variable: Panel B: *Index of Economic Development*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	0.205	−0.143	−0.277	−0.285	−0.223
	(0.248)	(0.198)	(0.174)	(0.174)	(0.181)
	[0.309]	[0.192]	[0.164]	[0.164]	[0.150]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	2,277	2,277	2,277	2,277	2,277
Adjusted R2	-0.000	0.379	0.557	0.560	0.573
Mean Dep. Var.	-0.085	-0.085	-0.085	-0.085	-0.085
SD Dep. Var.	4.051	4.051	4.051	4.051	4.051

Notes: Data is from the 2010 Mexico Population Census. Observations are municipalities in Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Index of Economic Development* is the first principal component index for a number of development outcomes in the census for a municipality (see Data Appendix). *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar date of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D6: Municipality Characteristics and Undetermined Festival Date

	Festival Date Determined			*Festival Date Undetermined*			Regression Estimates: *Festival Date Undetermined*		
	Obs.	Mean	SE	Obs.	Mean	SE	Coef.	Robust SE	Conley SE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Geographic Characteristics:									
Precipitation	1593	95.43	(1.15)	46	101.68	(7.93)	1.98	(6.81)	[5.45]
Temperature	1593	19.12	(0.10)	46	20.10	(0.61)	0.76	(0.49)	[0.53]
Land Suitability	1593	86.26	(0.33)	46	80.83	(2.53)	-4.84	(2.10)	[1.80]
Maize Suitability	1593	34.54	(0.55)	46	36.85	(2.96)	2.34	(2.98)	[3.02]
Area	1593	329.73	(12.57)	46	416.61	(81.26)	52.42	(80.22)	[80.17]
Longitude	1593	-98.31	(0.05)	46	-98.10	(0.28)	0.18	(0.11)	[0.09]
Latitude	1593	18.67	(0.04)	46	19.14	(0.23)	-0.11	(0.12)	[0.11]
Log(Dist. to Mexico City)	1593	5.43	(0.02)	46	5.41	(0.10)	0.07	(0.05)	[0.03]
Slope	1593	10.26	(0.16)	46	8.71	(1.00)	-0.82	(0.97)	[1.06]
Elevation	1593	1559.80	(19.68)	46	1314.65	(134.19)	-137.09	(94.00)	[99.79]
Colonial Characteristics:									
Has Colonial Characteristics (%)	1593	85.69	(0.88)	46	82.61	(5.65)	-0.94	(4.84)	[4.61]
Drought in 1545 (%)	1365	99.49	(0.19)	38	100	(0.00)	0.68	(0.69)	[0.55]
Log(Pop. Density in 1570)	1365	0.52	(0.03)	38	0.53	(0.21)	-0.12	(0.12)	[0.10]

Notes: Observations are municipalities in the *New Spain* region of Mexico.Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. See Data Appendix for more information on variables. Sample is limited to the *New Spain* region of Mexico. *Festival Date Undetermined* is an indicator variable equal to 1 if we were unable to determine the patron saint day festival in a municipality, and 0 otherwise. The value displayed for regressions estimates is the coefficient estimate for *Festival Date Undetermined*, conditional on state fixed effects, planting-month and harvest-month fixed effects, and festival week fixed effects. Robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D7: Municipality Characteristics and "Local" Patron Saints

	Official Patron Saint			*Local Patron Saint*			Regression Estimates: *Local Patron Saint*		
	Obs.	Mean	SE	Obs.	Mean	SE	Coef.	Robust SE	Conley SE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Geographic Characteristics:									
Precipitation	1593	95.43	(1.15)	32	78.25	(5.54)	-7.27	(5.67)	[4.96]
Temperature	1593	19.12	(0.10)	32	18.74	(0.72)	0.02	(0.38)	[0.33]
Land Suitability	1593	86.26	(0.33)	32	86.52	(3.09)	-0.53	(1.67)	[1.62]
Maize Suitability	1593	34.54	(0.55)	32	44.75	(3.17)	2.07	(2.56)	[2.71]
Area	1593	329.73	(12.57)	32	381.08	(74.68)	-42.10	(52.41)	[56.75]
Longitude	1593	-98.31	(0.05)	32	-99.92	(0.39)	-0.16	(0.10)	[0.10]
Latitude	1593	18.67	(0.04)	32	19.70	(0.18)	0.12	(0.10)	[0.11]
Log(Dist. to Mexico City)	1593	5.43	(0.02)	32	5.19	(0.15)	0.08	(0.05)	[0.05]
Slope	1593	10.26	(0.16)	32	6.98	(0.65)	-0.98	(0.66)	[0.67]
Elevation	1593	1559.80	(19.68)	32	1700.77	(138.01)	47.58	(76.38)	[67.00]
Colonial Characteristics:									
Has Colonial Characteristics (%)	1593	85.69	(0.88)	32	93.75	(4.35)	5.68	(4.55)	[4.55]
Drought in 1545 (%)	1365	99.49	(0.19)	30	100	(0.00)	0.98	(0.80)	[0.82]
Log(Pop. Density in 1570)	1365	0.52	(0.03)	30	0.56	(0.24)	-0.09	(0.11)	[0.09]

Notes: Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. See Data Appendix for more information on variables. *Local Patron Saint* is an indicator variable equal to 1 if the patron saint in a municipality is not an official Vatican patron saint, and 0 otherwise. The value displayed for regressions estimates is the coefficient estimate for *Local Patron Saint*, conditional on state fixed effects, planting-month and harvest-month fixed effects, and festival week fixed effects. Robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D8: Robustness to Missing Festival Dates: Development Outcomes
(Assuming All Municipalities with Missing Festival Dates Have Coinciding Festivals)

	Dependent Variable: Panel A: *Log HH Income*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.129	−0.107	−0.151	−0.151	−0.130
	(0.083)	(0.070)	(0.059)	(0.059)	(0.064)
	[0.087]	[0.063]	[0.051]	[0.049]	[0.051]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,639	1,639	1,639	1,639	1,632
Adjusted R2	0.001	0.343	0.536	0.542	0.559
Mean Dep. Var.	2.798	2.798	2.798	2.798	2.799
SD Dep. Var.	1.226	1.226	1.226	1.226	1.227

	Dependent Variable: Panel B: *Index of Economic Development*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.384	−0.213	−0.383	−0.394	−0.431
	(0.281)	(0.236)	(0.202)	(0.202)	(0.214)
	[0.293]	[0.222]	[0.178]	[0.178]	[0.171]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,639	1,639	1,639	1,639	1,632
Adjusted R2	0.000	0.343	0.563	0.568	0.587
Mean Dep. Var.	-0.577	-0.577	-0.577	-0.577	-0.575
SD Dep. Var.	4.039	4.039	4.039	4.039	4.047

Notes: Regressions in this table are identical to those in Table 2, except that municipalities with missing festival dates (previously not included in sample) are now included in the sample, and we assume that their festivals *all* coincide with planting or harvest (for these observations, festival week is randomly chosen for the inclusion of Festival-Week Fixed Effects in column 5). Data is from the 2010 Mexico Population Census. Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Index of Economic Development* is the first principal component index for a number of development outcomes in the census for a municipality (see Data Appendix). *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. For this table, we assume all undetermined festival dates are coinciding festivals. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar date of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D9: Robustness to Missing Festival Dates: Development Outcomes
(Assuming No Municipalities with Missing Festival Dates Have Coinciding Festivals)

	Dependent Variable: Panel A: *Log HH Income*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.214	−0.137	−0.178	−0.183	−0.157
	(0.091)	(0.077)	(0.066)	(0.065)	(0.069)
	[0.097]	[0.065]	[0.058]	[0.056]	[0.059]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,639	1,639	1,639	1,639	1,631
Adjusted R2	0.003	0.343	0.537	0.542	0.558
Mean Dep. Var.	2.798	2.798	2.798	2.798	2.798
SD Dep. Var.	1.226	1.226	1.226	1.226	1.227

	Dependent Variable: Panel B: *Index of Economic Development*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.594	−0.290	−0.471	−0.496	−0.508
	(0.306)	(0.253)	(0.219)	(0.218)	(0.234)
	[0.344]	[0.238]	[0.192]	[0.189]	[0.183]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,639	1,639	1,639	1,639	1,631
Adjusted R2	0.002	0.343	0.563	0.568	0.587
Mean Dep. Var.	-0.577	-0.577	-0.577	-0.577	-0.579
SD Dep. Var.	4.039	4.039	4.039	4.039	4.045

Notes: Regressions in this table are identical to those in Table 2, except that municipalities with missing festival dates (previously not included in sample) are now included in the sample, and we assume that *none* of their festivals coincide with planting or harvest (for these observations, festival week randomly chosen among non-planting and non-harvest months for inclusion of Festival-Week Fixed Effects in column 5). Data is from the 2010 Mexico Population Census. Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Index of Economic Development* is the first principal component index for a number of development outcomes in the census for a municipality (see Data Appendix). *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. For this table, we assume all undetermined festival dates are not coinciding festivals. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar date of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D10: Robustness to Including "Local" Patron Saints: Development Outcomes

	Dependent Variable: Panel A: *Log HH Income*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.187	−0.111	−0.160	−0.165	−0.121
	(0.091)	(0.076)	(0.066)	(0.065)	(0.070)
	[0.098]	[0.064]	[0.057]	[0.055]	[0.055]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,625	1,625	1,625	1,625	1,625
Adjusted R2	0.002	0.345	0.535	0.541	0.561
Mean Dep. Var.	2.809	2.809	2.809	2.809	2.809
SD Dep. Var.	1.227	1.227	1.227	1.227	1.227

	Dependent Variable: Panel B: *Index of Economic Development*				
	(1)	(2)	(3)	(4)	(5)
Festival Coincides with Maize Planting or Harvest	−0.514	−0.197	−0.406	−0.430	−0.372
	(0.307)	(0.255)	(0.221)	(0.219)	(0.230)
	[0.353]	[0.245]	[0.198]	[0.196]	[0.173]
State Fixed Effects	N	Y	Y	Y	Y
Geography Controls	N	N	Y	Y	Y
Colonial Controls	N	N	N	Y	Y
Planting-Month Fixed Effects	N	N	N	N	Y
Harvest-Month Fixed Effects	N	N	N	N	Y
Festival-Week Fixed Effects	N	N	N	N	Y
Observations	1,625	1,625	1,625	1,625	1,625
Adjusted R2	0.001	0.348	0.564	0.570	0.593
Mean Dep. Var.	-0.530	-0.530	-0.530	-0.530	-0.530
SD Dep. Var.	4.048	4.048	4.048	4.048	4.048

Notes: Regressions in this table are identical to those in Table 2, except that municipalities celebrating "local" saints (those not appearing in official Roman Catholic records outside of Mexico, previously not included in sample) are now included in the sample, and we use their actual festival celebration dates to determine whether they coincide with planting or harvest. Data are from the 2010 Mexico Population Census. Observations are municipalities in the *New Spain* region of Mexico. Robust standard errors are presented in parentheses and Conley (1999) standard errors calculated using a 100 km cut-off window are presented in brackets. *Index of Economic Development* is the first principal component index for a number of development outcomes in the census for a municipality (see Data Appendix). *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. For this table, we assume all undetermined festival dates are not coinciding festivals. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar date of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D11: Impact of Agriculturally-Coinciding Festivals on Migration Outcomes

	Dependent Variables:			
	% Born in a Different State	*% Different Municipality 5 Years Ago*	*% Different State 5 Years Ago*	*% Different Country 5 Years Ago*
	(1)	(2)	(3)	(4)
Festival Coincides with Maize Planting or Harvest	-0.006	0.000	-0.003	0.004
	(0.006)	(0.002)	(0.001)	(0.001)
	[0.005]	[0.002]	[0.001]	[0.001]
State Fixed Effects	Y	Y	Y	Y
Geography Controls	Y	Y	Y	Y
Colonial Controls	Y	Y	Y	Y
Planting-Month Fixed Effects	Y	Y	Y	Y
Harvest-Month Fixed Effects	Y	Y	Y	Y
Festival-Week Fixed Effects	Y	Y	Y	Y
Observations	1,593	1,593	1,593	1,593
Adjusted R2	0.546	0.198	0.199	0.299
Mean Dep. Var.	0.085	0.023	0.025	0.018
SD Dep. Var.	0.103	0.029	0.022	0.015

Notes: Data is from the 2010 Population Census. Observations are municipalities in the *New Spain* region of Mexico. Standard errors are clustered at the municipality level. *% Born in a Different State* is the share of individuals in a municipality that report being born in a different state than their current state of residence. *% Different Municipality 5 Years Ago* is the share of individuals in a municipality who report having lived in a different municipality (but within the same state) five years ago. *% Different State 5 Years Ago* is the share of individuals in a municipality who report having lived in a different state five years ago. *% Different Country 5 Years Ago* is the share of individuals in a municipality who report having lived abroad five years ago. *Festival Coincides with Maize Planting or Harvest* is an indicator variable equal to 1 if the saint day festival in a municipality occurs either 0 to 30 days prior to the optimal maize planting date or 0 to 30 days after the optimal maize harvest date for a municipality using FAO GAEZ data. *Geography Controls* includes mean temperature, mean precipitation, mean land suitability, the surface area, centroid latitude, centroid longitude, mean elevation, mean slope, log distance to Mexico City, and mean maize suitability for the municipality. *Colonial Controls* includes drought intensity in 1545 and log population density in 1570 using data from Sellers and Alix-Garcia (2018). For these colonial controls, values for municipalities with missing information are set to zero, and we control for an indicator variable equal to 1 if the municipality is not missing these colonial characteristics. *Planting & Harvest Month Fixed Effects* includes fixed effects for the optimal planting-month and harvest-month for maize for each municipality according to FAO GAEZ data. *Festival-Week Fixed Effects* are fixed effects for the calendar week of the municipality's saint day festival. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.